SGA 4, Exposé VI

Finiteness Conditions. Fibred Topoi and Sites. Applications to Questions of Passage to the Limit

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3. Finiteness conditions for a morphism of topoi

Definition 3.1. Let $f : E' \to E$ be a morphism of topoi. We say that f is *quasi-compact* (respectively *quasi-separated*) if, for every quasi-compact (respectively quasi-separated) object X of E, the object $f^*(X)$ is quasi-compact (respectively quasi-separated). We say that f is *coherent* if it is both quasi-compact and quasi-separated.

3.1.1

If f has one of the preceding properties, then every morphism of topoi isomorphic to f has the same property. It is also immediate that the composite of two quasi-compact (respectively quasi-separated, respectively coherent) morphisms of topoi is again quasi-compact (respectively quasi-separated, respectively coherent).

Proposition 3.2. With the notation of 3.1, let $(X_i)_{i \in I}$ be a generating family of quasi-compact (respectively coherent) objects of E. In the respective coherent case, assume that E' admits a generating family consisting of quasi-compact objects. Then f is quasi-compact (respectively coherent) if and only if, for every $i \in I$, the object $f^*(X_i)$ is quasi-compact (respectively coherent).

The necessity is trivial. For sufficiency in the quasi-compact case, let X be a quasi-compact object of E. There is a finite epimorphic family of morphisms $X_i \to X$. Pulling back, one obtains a finite epimorphic family $f^*(X_i) \to f^*(X)$, with the $f^*(X_i)$ quasi-compact by hypothesis; hence $f^*(X)$ is quasi-compact.

In the coherent case, it remains to prove quasi-separatedness. If X is quasi-separated in E, this may be expressed by the existence of an epimorphic family $X_i \to X$ such that the fibre products $X_i \times_X X_j$ are quasi-compact. After applying f^*, we get a covering family $f^*(X_i) \to f^*(X)$ such that the fibre products

$$f^*(X_i) \times_{f^*(X)} f^*(X_j)$$

are quasi-compact, since by the quasi-compact part already proved f^* sends quasi-compact objects to quasi-compact objects. Since the $f^*(X_i)$ are coherent, one concludes again by the criterion 1.17.

Corollary 3.3. Let C and C' be two $\mathcal{U}$-sites in which fibre products are representable and every covering family admits a finite covering subfamily. Let $g : C \to C'$ be a morphism of sites. Then the morphism of topoi

$$f : C^\sim \to C'^\sim$$

defined by g is coherent.

Indeed, $\varepsilon_C(C)$ (respectively $\varepsilon_{C'}(C')$) is a generating family of $C^\sim$ (respectively $C'^\sim$) satisfying the corresponding hypotheses of 3.2.

Proposition 3.4. Let E be a locally coherent topos, and let $f : X \to Y$ be an arrow of E, whence a morphism of induced topoi $E_{/X} \to E_{/Y}$. This morphism of topoi is quasi-compact (respectively quasi-separated, respectively coherent) if and only if the arrow f is quasi-compact (respectively quasi-separated, respectively coherent).

The quasi-compact case follows directly from the definitions, without any hypothesis on E. The quasi-separated case is precisely 2.8, and the coherent case is the conjunction of the two preceding cases.

Proposition 3.5. Let $f : E' \to E$ be a morphism of locally coherent topoi. Let $(X_i)_{i \in I}$ be a generating family in E consisting of coherent algebraic objects. The following conditions are equivalent.

(i) For every quasi-compact arrow u of E, the arrow $f^*(u)$ is quasi-compact.

(i bis) For every arrow $u : X_i \to X_j$, the arrow $f^*(u)$ is quasi-compact.

(ii) For every coherent object Y' of E', every coherent algebraic object Y of E, and every arrow $v : Y' \to f^*(Y)$, the corresponding morphism of topoi

$$f_v : E'_{/Y'} \to E_{/Y}$$

obtained by composing the localization morphism $E'_{/Y'} \to E'_{/f^*(Y)}$ deduced from v with the morphism $E'_{/f^*(Y)} \to E_{/Y}$ induced by f, is coherent.

(ii bis) The same condition as in (ii), but restricted to a set of data $v_\alpha : Y'_\alpha \to f^*(Y_\alpha)$ such that the Y'_α are algebraic and cover the final object of E'.

(ii ter) Given a generating family $(X'_{i'})_{i' \in I'}$ of E' consisting of coherent objects, and, for every $i' \in I'$, an index $i \in I$ and an arrow $v_{i'} : X'_{i'} \to f^*(X_i)$, the morphism of topoi $E'_{/X'_{i'}} \to E_{/X_i}$ defined by $v_{i'}$ is coherent for every i'.

Since every arrow $X_i \to X_j$ is quasi-compact, (i) implies (i bis). Conversely assume (i bis) and prove (i). Let $u : X \to Y$ be a quasi-compact arrow of E. In terms of an epimorphic family $X_i \to Y$, the hypothesis on u says that the objects $X \times_Y X_i$ are quasi-compact; equivalently, for every i there is a finite epimorphic family of arrows $X_j \to X \times_Y X_i$. Pulling back by f^*, we get an epimorphic family $f^*(X_i) \to f^*(Y)$, and, for every i, a finite epimorphic family

$$f^*(X_j) \to f^*(X \times_Y X_i) \simeq f^*(X) \times_{f^*(Y)} f^*(X_i).$$

The composite with the second projection to $f^*(X_i)$ is quasi-compact by (i bis). Hence the projection

$$f^*(X) \times_{f^*(Y)} f^*(X_i) \to f^*(X_i)$$

is quasi-compact for every i, and consequently $f^*(u)$ is quasi-compact.

It remains to prove (i) $\Rightarrow$ (ii) and (ii bis) $\Rightarrow$ (i), since (ii) $\Rightarrow$ (ii bis) is trivial and (ii ter) is a special case of (ii bis). The implication (i) $\Rightarrow$ (ii) is immediate: an object X of $E_{/Y}$ is quasi-compact (respectively quasi-separated) if and only if its structural morphism $X \to Y$ (respectively its diagonal $X \to X \times_Y X$) is quasi-compact, and the same criterion applies after pulling back by f. Finally assume (ii bis), and let $u : X \to Y$ be quasi-compact in E. Since the Y'_α cover the final object of E', it is enough to prove that the restrictions $f^*(u)|Y'_\alpha$ are quasi-compact. These restrictions are precisely $f^*_{v_\alpha}(u|Y_\alpha)$, reducing the assertion to the following corollary.

Corollary 3.6. Let $f : E' \to E$ be a coherent morphism of locally coherent topoi. Then, for every quasi-compact arrow u of E, the arrow $f^*(u)$ is quasi-compact.

By the implication (i bis) $\Rightarrow$ (i) already proved, it is enough to prove this when $u : X \to Y$ has X and Y coherent. But this follows at once from the hypothesis on f, which implies that $f^*(X)$ and $f^*(Y)$ are coherent.

Definition 3.7. Let $f : E' \to E$ be a morphism of locally coherent topoi. We say that f is *locally coherent* if it satisfies the equivalent conditions of 3.5.

Here "locally" means locally upstairs.

3.7.1

This notion again depends only on the isomorphism class of the morphism of topoi f, and is manifestly stable under composition of morphisms of topoi. Moreover, by 3.5, every coherent morphism is locally coherent; conversely, if E' and E are coherent, every locally coherent morphism is coherent. Criterion 3.5(ii bis) immediately gives the following criterion.

Corollary 3.7. Let $f : E' \to E$ be a morphism of locally coherent topoi. Let $(Y_i)_{i\in I}$ and $(Y_i')_{i\in I}$ be families of objects of E and E', and for each i let $v_i : Y_i' \to f^*(Y_i)$, whence a morphism of topoi

$$f_{v_i} : E'_{/Y_i'} \to E_{/Y_i}.$$

Assume that the Y_i' cover the final object of E'. Then f is locally coherent if and only if each f_{v_i} is locally coherent.

Applying this to the identity morphism of an induced topos gives:

Corollary 3.8. Let E be a locally coherent topos and let $v : X \to Y$ be an arrow of E. Then the morphism of induced topoi $E_{/X} \to E_{/Y}$ defined by v is locally coherent.

Remark 3.9. It seems that all algebraic morphisms of topoi encountered in practice are locally coherent; for this reason the term "locally coherent" is probably not destined for very frequent use. Equivalently, the coherent morphisms of topoi encountered in practice are coherent. Nevertheless one can construct non-coherent morphisms between coherent topoi, and in particular a non-coherent morphism from the punctual topos to a noetherian topos E; cf. 2.17(i).

Examples 3.10: topoi associated with schemes. Return to Example 1.22, the category (Sch) with one of the topologies T_i of SGA IV 6.3. For every scheme X, let X_{T_i} be the $\mathcal{V}$-topos defined by the induced site $(\mathrm{Sch})_{/X}$. A morphism of schemes $f : X \to Y$ defines a morphism of topoi $f_{T_i} : X_{T_i} \to Y_{T_i}$, and by 3.4 and 1.22 this morphism is quasi-compact (respectively quasi-separated, respectively coherent) if and only if the morphism of schemes f is quasi-compact (respectively quasi-separated, respectively coherent) in the usual sense of EGA IV. In any case, f_{T_i} is locally coherent by 3.8.

One may also associate with a scheme X the $\mathcal{U}$-topoi $\mathrm{Top}(X)$ and $\mathrm{Top}(X_{\mathrm{et}})$, as in 1.22.1; to every morphism of schemes $f : X \to Y$ one then associates morphisms $\mathrm{Top}(f)$ and $\mathrm{Top}(f_{\mathrm{et}})$ of the corresponding topoi. Let $T(f)$ be one of these two morphisms. By 3.2 one verifies immediately that $T(f)$ is always locally coherent, and that it is quasi-compact (respectively quasi-separated, respectively coherent) if and only if f is quasi-compact (respectively quasi-separated, respectively coherent) in the usual sense.

These observations show again that the terminology introduced here is compatible with the accepted terminology in scheme theory, and therefore deserves acceptance even from the most reluctant reader.

Exercise 3.11: coherent topoi and pretopoi.

a) A $\mathcal{U}$-*pretopos* (or simply pretopos) is a category C satisfying the following conditions:

   1) finite projective limits in C are representable;
   2) finite sums in C are representable, disjoint, and universal;
   3) equivalence relations in C are effective, and every epimorphism in C is universally effective;
   4) C is equivalent to a category belonging to $\mathcal{U}$.

   Show that if E is a coherent $\mathcal{U}$-topos, then the full subcategory E_{coh} of coherent objects is a $\mathcal{U}$-pretopos, and the inclusion $E_{\text{coh}} \to E$ is left exact, commutes with finite sums, and commutes with passage to quotients by equivalence relations. Show that the topology induced by E on $C = E_{\text{coh}}$ is the *precanonical topology*, namely the topology whose covering families $X_i \to X$ are those which admit a finite subfamily covering for the canonical topology of C. Hence E can be reconstructed, up to equivalence, from the pretopos $C = E_{\text{coh}}$ as the topos $C^{\sim}$, where C is endowed with its precanonical topology.

b) Let C be a $\mathcal{U}$-pretopos, endowed with its precanonical topology. Show that every morphism f in C factors as $f''f'$, with f' an epimorphism and f'' a monomorphism. Show that $E = C^{\sim}$ is coherent, and that the canonical functor $\varepsilon : C \to E$ induces an equivalence of C with E_{coh}. First show that ε is fully faithful, left exact, and commutes with finite sums and quotients by equivalence relations, and then show that a coherent subobject in E of an object of C lies in the essential image of C. Thus a $\mathcal{U}$-pretopos C can be recovered, up to equivalence of categories, from its associated $\mathcal{U}$-topos $E = C^{\sim}$.

c) Let C and C' be $\mathcal{U}$-pretopoi. Show that a functor $\varphi : C' \to C$ is a morphism of sites from C to C', for the precanonical topologies, if and only if φ is left exact and commutes with finite sums and quotients by equivalence relations.

d) With the notation of c), assume that C and C' are associated with two coherent $\mathcal{U}$-topoi E and E' as in a). Show that the canonical functor

$$\mathrm{MorSite}(C, C') \to \mathcal{H}om_{\text{top}}(E, E')$$

   induces an equivalence from the first category, described in c), onto the full subcategory of the second consisting of coherent morphisms of topoi $E \to E'$. A quasi-inverse sends a coherent morphism $f : E \to E'$ to the functor $E'_{\text{coh}} = C' \to E_{\text{coh}} = C$ induced by f^*.

e) Let C be a $\mathcal{U}$-pretopos and $E = C^{\sim}$. Express directly, in exactness properties of C, the equivalent conditions of 1.25. Show that an object X of C is noetherian in E if and only if every increasing sequence of subobjects of X in C is stationary; hence E is noetherian if and only if every object of C satisfies this condition.

Exercise 3.12. A topos E is called *finite* if there exists a finite category C such that E is equivalent to $\widehat{C}$.

a) Show that E is finite if and only if E has enough essential points and the category $\mathrm{Point}_{\text{ess}}(E)$ of essential points of E is equivalent to a finite category.

b) Assume E finite. Prove that every point of E is essential. Prove that the 2-functors $E \mapsto \mathrm{Point}(E)$ and $C \mapsto \widehat{C}$ establish 2-equivalences between the 2-category of finite topoi and the 2-category of Karoubian categories equivalent to finite categories. Prove also that every morphism of finite topoi is essential.

c) Assume $E = \widehat{C}$, with C equivalent to a finite category, and let F be a coherent topos. Recall that morphisms of topoi $f : E \to F$ correspond to functors $u : C \to \mathrm{Point}(F)$. Show that f is coherent if and only if it sends coherent points of E to coherent points of F, equivalently if $u(X)$ is a coherent point of F for every $X \in \mathrm{ob}\, C$. Deduce that every morphism from a finite topos to a finite topos is coherent.

d) Let X be a sober topological space. Show that $\mathrm{Top}(X)$ is finite if and only if X is a finite set.

4. Finiteness conditions in a topos obtained by gluing

The present section will not be used later in the Seminar.

4.1

Let E be a topos and U an open of E, that is, a subobject of the final object e of E. Consider the corresponding open and closed subtopoi of E:

$$E' = E_{/U}, \qquad E'' = E_{/U^c}.$$

We denote by

$$j : E' \to E, \qquad i : E'' \to E$$

the canonical morphisms of topoi. Recall that j is quasi-compact (respectively coherent) if and only if the inclusion $j : U \to e$ is a quasi-compact (respectively coherent) morphism in E. On the other hand, $j : E' \to E$ is always quasi-separated, because $j : U \to e$ is so. We shall give criteria for the closed subtopos E'' to be coherent and for the inclusion $i : E'' \to E$ to be coherent.

Proposition 4.2. With the notation of 4.1, the inclusion morphism $i : E'' \to E$ is quasi-compact; that is, for every quasi-compact object X of E, the object $i^*(X)$ of E'' is quasi-compact. Moreover, if X is pre-noetherian, then $i^*(X)$ is pre-noetherian.

This follows from Definition 1.1 and the following lemma.

Lemma 4.2.1. The map $Y \mapsto i^*(Y)$ induces an isomorphism of ordered sets between the set of subobjects of X containing the subobject $X_U = X \times U$ of X, and the set of subobjects of $i^*(X)$.

Indeed, since i_* is conservative (even fully faithful) and left exact, a morphism $u : Y \to Z$ in E'' is a monomorphism if and only if $i_*(u)$ is. Identifying E'', through i_*, with the full subcategory of E described in IV 9.7, subobjects in E'' of an object Z are the subobjects Y of Z in E which still belong to E'', equivalently which contain the copy of U. For $Z = i_* i^*(X) = X_{\complement U}$, described by the pushout

$$X_{\complement U} = X \amalg_{X_U} U,$$

a subobject of Z is the same as a pair consisting of a subobject $Y_1 \subset X$ and a subobject $Y_2 \subset U$, with equal inverse images in X_U. The condition that this subobject contain U says $Y_2 = U$, and the remaining condition on Y_1 is precisely $X_U \subset Y_1$. This proves the claimed order isomorphism.

Corollary 4.3.

a) Let X be an object of E. To prove X quasi-compact it suffices that $i^*(X)$ and $j^*(X)$ be quasi-compact; this condition is also necessary if $j : U \to e$ is quasi-compact.

b) Let Y be an object of E''. To prove Y quasi-compact it suffices that $i_*(Y)$ be quasi-compact; this condition is also necessary if U is quasi-compact.

For a), sufficiency follows from Definition 1.1 and the conservativity of the pair (i^*, j^*); necessity follows from the quasi-compactness of i and j. For b), since $Y \simeq i^*i_*(Y)$, sufficiency follows from 4.2. Necessity follows from a), using $i^*i_*(Y) \simeq Y$ and $j^*i_*(Y) \simeq U$.

Corollary 4.4. Let Y be an object of E''. If U is quasi-compact, or if E admits a generating family consisting of quasi-compact objects, then for Y to be quasi-separated (respectively coherent) it suffices that $i_*(Y)$ be quasi-separated (respectively coherent). If U is quasi-separated (respectively coherent) and $j : U \to e$ is quasi-compact, then for Y to be quasi-separated (respectively coherent) it is necessary that $i_*(Y)$ be so.

The coherent case follows from the quasi-separated case together with 4.3(b). Suppose that $i_*(Y)$ is quasi-separated. We must prove that if Y' and Y'' are quasi-compact objects over Y, then $Y' \times_Y Y''$ is quasi-compact. If U is quasi-compact, it suffices, by 4.3(b), to prove that

$$i_*(Y' \times_Y Y'')$$

is quasi-compact; this follows because $i_*(Y')$ and $i_*(Y'')$ are quasi-compact, because i_* commutes with fibre products, and because $i_*(Y)$ is quasi-separated. If instead E has a generating family of quasi-compact objects, write $i_*(Y')$ as a filtered inductive limit of its quasi-compact subobjects X'_α. Then $Y' \simeq i^*i_*(Y')$ is a filtered inductive limit of the $i^*(X'_\alpha)$, hence, since Y' is quasi-compact, is equal to one of them. Similarly $Y'' = i^*(X''_\beta)$. Thus

$$Y' \times_Y Y'' = i^*(X'_\alpha \times_{i_*(Y)} X''_\beta),$$

which is quasi-compact by 4.2.

Conversely, assume U quasi-separated and $j : U \to e$ quasi-compact. If Y is quasi-separated, then $i_*(Y)$ is quasi-separated: for quasi-compact objects X', X'' over $i_*(Y)$, it suffices by 4.3(a) to check that the images under i^* and j^* of $X' \times_{i_*(Y)} X''$ are quasi-compact. The first assertion follows from 4.2 and the quasi-separatedness of Y; the second image is $j^*(X') \times j^*(X'')$, which is quasi-compact because j is quasi-compact and U is quasi-separated.

Corollary 4.5. If E admits a generating subcategory consisting of quasi-compact (respectively pre-noetherian) objects, then so does E''.

This follows from 4.2 and the following lemma.

Lemma 4.5.1. If $(X_\alpha)_\alpha$ is a generating family in E, then $(i^*(X_\alpha))_\alpha$ is a generating family in E''.

Indeed the family of functors $Y \mapsto \mathrm{Hom}(i^*(X_\alpha), Y)$ on E'' is conservative, since $\mathrm{Hom}(i^*(X_\alpha), Y) \simeq \mathrm{Hom}(X_\alpha, i_*(Y))$ and i_* is conservative.

Proposition 4.6. With the notation of 4.1:

a) If E admits a generating family consisting of coherent objects, then the same is true for the closed subtopos E'', and the inclusion $i : E'' \to E$ is coherent.

b) Assume $j : U \to e$ quasi-compact. If E is locally coherent (respectively coherent, quasi-separated, locally noetherian, noetherian, locally perfect, perfect), then so is E'', and the inclusion $i : E'' \to E$ is coherent.

For a), by 4.5 the topos E'' has a generating family of quasi-compact objects, so by 3.2 it is enough to show that $i^*(X)$ is coherent for every coherent X of E. Using compatibility of the formation of the complement of an open with localization, one reduces to the case where X is the final object of E. Then it is enough to prove that the final object of E'' is coherent. By 4.4 this follows from the fact that $i_*(e_{E''}) = e_E$ is coherent.

For b), under each of the stated hypotheses E has a generating family of coherent objects, so a) already gives that i is coherent and that E''_{coh} generates E''. If E is coherent, the stability of E''_{coh} under finite projective limits follows from the criterion 4.4 (which applies since now U is coherent and $j : U \to e$ is quasi-compact), together with the left exactness of i_*. By localization, the locally coherent case follows. If E is quasi-separated, then the final object of E'' is quasi-separated by 4.4; products of quasi-separated objects in E'' are quasi-separated by the same criterion and the fact that i_* commutes with products.

Since a topos is locally noetherian (respectively noetherian) if and only if it is locally coherent (respectively coherent) and has a generating family consisting of pre-noetherian objects, 4.5 gives the noetherian assertions. If E is perfect, then E'' is already coherent, and it remains to check criterion 1.25(i): every object Y of E'' must be a filtered inductive limit of coherent objects. Since $i_*(Y)$ has such a presentation by coherent objects X_α, one has $Y \simeq i^* i_*(Y)$, a filtered inductive limit of the coherent objects $i^*(X_\alpha)$. Localizing gives the locally perfect case.

Remark 4.6.1. In fact in 4.6(b) the assumption that $j : U \to e$ be quasi-compact is unnecessary except perhaps in the quasi-separated case. It is enough, by the proof, to see that E coherent implies E'' coherent. Since U is a filtered inductive limit of its quasi-compact subobjects U_α, Section 7 will show that E'' identifies with the projective-limit topos of the closed subtopoi E''_α complementary to the $E_{/U_\alpha}$. These are coherent topoi with coherent transition morphisms; hence the limit E'' is coherent.

Corollary 4.7. Assume that E_{coh} is generating, and let X be an object of E.

a) For X to be quasi-separated it is necessary that $i^*(X)$ and $j^*(X)$ be quasi-separated; this condition is also sufficient if $j : U \to e$ is quasi-compact.

b) If $j : U \to e$ is quasi-compact, then X is coherent if and only if $i^*(X)$ and $j^*(X)$ are coherent.

c) If E is coherent and $j : U \to e$ is quasi-compact, then E is perfect if and only if E' and E'' are perfect.

Necessity in a) follows from the quasi-separatedness of j and of i, the latter by 4.6(a). For sufficiency, let X' and X'' be quasi-compact objects over X; it suffices by 4.3(a) to prove that the images under i^* and j^* of $X' \times_X X''$ are quasi-compact. This follows because i and j are quasi-compact and because $i^*(X)$, $j^*(X)$ are quasi-separated. Assertion b) is the conjunction of a) and 4.3(a). Necessity in c) was already proved in 4.6(b); sufficiency follows because, for a coherent topos, perfection is equivalent to stability of coherent objects under finite inductive limits, and b) gives this stability after gluing.

Exercise 4.8.

a) Resolve the following question, to which the writer confesses not to know the answer: with the notation of 4.1, if E is quasi-separated (respectively algebraic), is E'' so as well?

b) Assume that E is equivalent to a topos of the form $\widehat{C}$, where C is equivalent to a $\mathcal{U}$-small category. Prove directly, using 2.17(c) and IV 9.24, that if E is locally coherent (respectively coherent, algebraic, quasi-separated, locally noetherian, noetherian), then the same is true of E''.

4.9

Keep the notation of 4.1. Recall that the datum of a pair (E, U), consisting of a topos E and an open U of E, is essentially equivalent to the datum of a triple (E', E'', f), where E' and E'' are topoi and

$$f : E' \to E'' \tag{4.9.1}$$

is a *gluing functor*, that is, a left exact accessible functor. Starting with (E, U), the associated gluing functor is

$$f = i^* j_*. \tag{4.9.2}$$

We shall express, in terms of E', E'', and f, the fact that E is coherent (respectively perfect) and that U is quasi-compact, equivalently that E and E' are coherent. We already know that this implies E'' coherent. Thus the problem is the following: given two coherent topoi E', E'' and a gluing functor f, under what conditions is the glued topos E coherent (respectively perfect)? The perfect case reduces to the coherent case, since E is perfect if and only if E, E', and E'' have the corresponding coherent and perfect properties; however the final criterion takes a cleaner form when E' is already assumed perfect.

4.9.3

First suppose E coherent. Then the full subcategory E_{coh} of coherent objects $X = (X', X'', u : X'' \to f(X'))$ is reconstructed as the subcategory E_0 consisting of those X for which $X' = j^*(X)$ and $X'' = i^*(X)$ are coherent. A necessary condition for E to be coherent is therefore that this subcategory E_0 be generating. This condition is also sufficient: by 4.3(a), E_0 consists of quasi-compact objects; it is plainly stable under finite projective limits, and one concludes by 2.4.5.

Recall also that giving the gluing functor f is equivalent to giving a sheaf on E'' with values in $\mathrm{Pro}(E')^0$,

$$G \in \mathrm{Sh}(E'', \mathrm{Pro}(E')^0), \qquad G : E''^0 \to \mathrm{Pro}(E')^0, \tag{4.9.4}$$

or equivalently a functor

$$g : E'' \to \mathrm{Pro}(E') \tag{4.9.5}$$

which commutes with inductive limits. If C is a generating subcategory of E'', endowed with the induced topology, this datum is also equivalent to a sheaf on C with values in $\mathrm{Pro}(E')^0$,

$$G_C \in \mathrm{Sh}(C, \mathrm{Pro}(E')^0), \qquad G_C : C^0 \to \mathrm{Pro}(E')^0, \tag{4.9.6}$$

or to a functor

$$g_C : C \to \mathrm{Pro}(E') \tag{4.9.7}$$

satisfying the expected left exactness conditions. Of course g_C is just the restriction of g to C. The case most useful here is $C = E''_{\mathrm{coh}}$, giving

$$G_0 \in \mathrm{Sh}(E''_{\mathrm{coh}}, \mathrm{Pro}(E')^0), \qquad G_0 : {E''}^0_{\mathrm{coh}} \to \mathrm{Pro}(E')^0, \tag{4.9.8}$$

$$g_0 = G_0^0 : E''_{\mathrm{coh}} \to \mathrm{Pro}(E'). \tag{4.9.9}$$

Either of these data is again equivalent to the original gluing functor f.

4.9.10

Assume that the generating full subcategory C of E'' consists of quasi-compact objects and is stable in E'' under finite sums, quotients by equivalence relations, and fibre products. This is the case, for instance, for $C = E''_{\mathrm{coh}}$. If P is a category with finite projective limits, then a functor $G_C : C^0 \to P$ is a sheaf with values in P if and only if the opposite functor $g_C = G^0_C : C \to P^0$ commutes with finite sums and quotients by equivalence relations. This describes explicitly the sheaves 4.9.6, and hence the gluing functors $f : E' \to E''$.

Proposition 4.10. Let E' and E'' be coherent topoi, and let $f : E' \to E''$ be a gluing functor, that is, a left exact accessible functor. Let E be the topos obtained by gluing. Let E'_{coh} denote the full subcategory of coherent objects of E', and let E'_{PF} denote the full subcategory of objects X such that the covariant representable functor $\mathrm{Hom}(X, -)$ commutes with small filtered inductive limits. Consider the following conditions.

(i) E is coherent.

(ii) For every object X' of E', there is a family of morphisms $v_\alpha : Y'_\alpha \to X'$, with coherent sources, such that the family $f(v_\alpha) : f(Y'_\alpha) \to f(X')$ is covering in E''.

(ii bis) f commutes with small filtered inductive limits, and (ii) holds for every $X' \in E'_{PF}$.

(ii ter) The canonical functor

$$\pi : \mathrm{Point}(E'') \to \mathrm{Pro}(E') \tag{4.10.1}$$

defined by f factors, up to isomorphism, through $\mathrm{Pro}(E'_{\mathrm{coh}})$:

$$\pi_0 : \mathrm{Point}(E'') \to \mathrm{Pro}(E'_{\mathrm{coh}}). \tag{4.10.2}$$

(iii) f commutes with small filtered inductive limits.

(iii bis) The functor g_0 in 4.9.9 factors, up to isomorphism, through a functor

$$E''_{\mathrm{coh}} \to \mathrm{Pro}(E'_{PF}) \tag{4.10.3}$$

via the fully faithful functor $\mathrm{Pro}(E'_{PF}) \to \mathrm{Pro}(E')$.

(iii ter) The canonical functor π in 4.10.1 factors, up to isomorphism, as

$$\pi_1 : \mathrm{Point}(E'') \to \mathrm{Pro}(E'_{PF}). \tag{4.10.4}$$

(iv) The functor g_0 in 4.9.9 factors, up to isomorphism, through a functor

$$E''_{\mathrm{coh}} \to \mathrm{Pro}(E'_{\mathrm{coh}}). \tag{4.10.5}$$

Then one has the diagram of implications

$$\text{(iv)} \Rightarrow \text{(i)} \Leftrightarrow \text{(ii)} \Leftrightarrow \text{(ii bis)} \Leftrightarrow \text{(ii ter)} \Rightarrow \text{(iii)} \Leftrightarrow \text{(iii bis)} \Leftrightarrow \text{(iii ter)}. \tag{4.10.6}$$

Corollary 4.11.

a) If E' is perfect, then all the conditions in 4.10 are equivalent. In particular, E is coherent if and only if f commutes with filtered inductive limits, or equivalently if and only if g_0 in 4.9.9 factors, up to isomorphism, through $\mathrm{Pro}(E'_{\mathrm{coh}})$.

b) The topos E is perfect if and only if E' and E'' are perfect and f (respectively g_0) satisfies the condition in a).

Indeed, a) follows from the fact that E' perfect means $E'_{\mathrm{coh}} = E'_{PF}$, so that (iii bis) implies (iv). Assertion b) follows from the preliminary remarks in 4.9.

4.12. Proof of 4.10

a) Unwinding the equivalent condition to (i) obtained in 4.9.3, the full subcategory E_0 of E must be generating. Thus, for every object

$$X = (X', X'', u : X'' \to f(X'))$$

of E, the family of all morphisms

$$v = (v', v'') : Y = (Y', Y'', v : Y'' \to f(Y')) \to X,$$

with $Y \in E_0$, that is with Y' and Y'' coherent, must be epimorphic. Since (i^*, j^*) is conservative, this means that the corresponding family $Y' \to X'$ is epimorphic in E', and the family $Y'' \to X''$ is epimorphic in E''. The first assertion is clear by taking objects Y lying over U, that is, with $Y'' = \varnothing$, because E'_{coh} generates E'. It remains to express the second assertion for arbitrary X.

b) Assume condition (ii) for the object X' considered. Then there is an epimorphic family $Y''_\beta \to X''$ whose composites with $u : X'' \to f(X')$ lift through morphisms $Y''_\beta \to f(Y'_{\varphi(\beta)})$. Since E''_{coh} generates E'', the Y''_β may be chosen coherent. The resulting commutative diagrams

$$\begin{tikzcd} Y''_\beta \arrow[r] \arrow[d] & f(Y'_{\varphi(\beta)}) \arrow[d] \\ X'' \arrow[r] & f(X') \end{tikzcd}$$

provide the desired epimorphic family in E. Thus (ii) implies (i). Conversely, assuming (i), apply the condition above to

$$X = j_*(X') = (X', f(X'), \mathrm{id} : f(X') \xrightarrow{\sim} f(X')).$$

Since the corresponding $Y'' \to f(X')$ form an epimorphic family, the family of the $f(v')$: $f(Y') \to f(X')$ is a fortiori epimorphic. Hence (i) and (ii) are equivalent.

c) Condition (iv) means that, for every object X'' of E''_{coh}, the pro-representable functor

$$g(X'') : X' \longmapsto \mathrm{Hom}(X'', f(X'))$$

on E' is pro-represented by a pro-object whose components lie in E'_{coh}. Equivalently, in the filtered category $E'_{/g(X'')}$ of pairs (X', u), where $u : X'' \to f(X')$, the full subcategory of those

(X'_α, u_α) with X'_α coherent is cofinal. Thus for every $u : X'' \to f(X')$ there is $v'_\alpha : X'_\alpha \to X'$ with X'_α coherent such that u factors through $f(v'_\alpha)$. It is then clear that (iv) implies (ii), since for fixed X' the family of all arrows $u : X'' \to f(X')$ with coherent source X'' is epimorphic. Hence (iv) implies (i).

d) Let p be a point of E''. By definition $\pi(p)$ is the pro-object of E' pro-representing the functor $h : X' \mapsto f(X')_p$. Saying that it comes from $\mathrm{Pro}(E'_{\mathrm{coh}})$ means that, in the category of pairs (X', u) with $u \in f(X')_p$, the subcategory with X' coherent is cofinal. Since a coherent topos has enough points, this condition, for all points p, is equivalent to the assertion that for every X' the family $f(X'_0) \to f(X')$, with X'_0 coherent, is epimorphic. This is exactly (ii). Thus (ii) is equivalent to (ii ter).

e) The condition that f commute with filtered inductive limits is equivalent, since the family of fibre functors of E'' is conservative, to the same condition for the composite functors $h : X' \mapsto f(X')_p$. This is equivalent to saying that the pro-object $\pi(p)$ pro-representing h belongs to the essential image of $\mathrm{Pro}(E'_{PF})$, by the following lemma.

Lemma 4.12.1. Let E' be a topos for which E'_{coh} is generating, and let $X' \in \mathrm{Ob}\,\mathrm{Pro}(E')$. Then X' belongs to the essential image of $\mathrm{Pro}(E'_{PF})$ if and only if the functor $h : E' \to \mathbf{Set}$ pro-represented by X' commutes with filtered inductive limits.

Since a filtered inductive limit of functors commuting with filtered inductive limits has the same property, one direction follows directly from the definition of E'_{PF}. Conversely, suppose that h commutes with filtered inductive limits. In the filtered category $E'_{/h}$ of pairs (X', u) with $u \in h(X')$, the full subcategory consisting of pairs with $X' \in E'_{PF}$ is cofinal: indeed every object X' is a filtered inductive limit of objects X'_i of E'_{PF}, and hence $h(X')$ is the filtered inductive limit of the $h(X'_i)$.

It remains only to prove (ii bis) $\Rightarrow$ (ii) and (iii) $\Leftrightarrow$ (iii bis). The first implication is immediate, since every object of E' is a filtered inductive limit of objects of E'_{PF}. For the second, the family of functors

$$Y'' \longmapsto \mathrm{Hom}(X'', Y''), \qquad X'' \in E''_{\mathrm{coh}},$$

is conservative and consists of functors commuting with filtered inductive limits. Hence f commutes with filtered inductive limits if and only if the composite functors $X' \mapsto \mathrm{Hom}(X'', f(X'))$ do so, which by Lemma 4.12.1 is equivalent to $g_0(X'')$ lying in the essential image of $\mathrm{Pro}(E'_{PF})$.

Remark 4.13. It is useful to spell out, in sheaf-theoretic language, condition 4.10(iv). Consider the commutative diagram of functors

$$\begin{tikzcd} \mathrm{Pro}(E'_{\mathrm{coh}})^0 \arrow[r, hook, "\alpha"] \arrow[d, "\beta_0"'] & \mathrm{Pro}(E')^0 \arrow[d, "\beta"] \\ (E'_{\mathrm{coh}})^\vee & (E')^\vee \arrow[l, "\gamma"] \end{tikzcd} \tag{4.13.1}$$

where the $\vee$'s denote categories of set-valued functors, and γ is restriction; the other arrows are the usual fully faithful functors. The inclusion α need not commute with projective limits, even finite ones, so a sheaf with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ may become only a presheaf after inclusion into $\mathrm{Pro}(E')^0$. Nevertheless, if a presheaf F with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ has αF a sheaf with values in $\mathrm{Pro}(E')^0$, then F itself is a sheaf. This follows because the sheaf condition is a left exactness property, and β, β_0, and γ commute with small projective limits.

Furthermore, from the characterization 4.9.10 of sheaves on E''_{coh} with values in an arbitrary category P, it follows that if $\alpha : P \to Q$ is such that $\alpha^0 : P^0 \to Q^0$ commutes with finite sums and quotients by equivalence relations, then composition with α sends sheaves to sheaves. Applying this to the inclusion $\mathrm{Pro}(E'_{\mathrm{coh}}) \to \mathrm{Pro}(E')$, one concludes that a functor

$$\varphi : {E''}^0_{\mathrm{coh}} \to \mathrm{Pro}(E'_{\mathrm{coh}})^0 \tag{4.13.2}$$

is a sheaf on E''_{coh} with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ if and only if $\alpha \circ \varphi$ is a sheaf with values in $\mathrm{Pro}(E')^0$. Thus the category of gluing functors $f : E' \to E''$ satisfying 4.10(iv) is canonically equivalent to the category of such sheaves φ. Such a sheaf canonically determines a gluing functor, up to unique isomorphism, and the topos E obtained by gluing is coherent. If E' is perfect, all coherent topoi obtained from E' and E'' by gluing arise in this way.

Since E'_{coh} is equivalent to a small category and is stable under finite projective limits, $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ identifies with the full subcategory of $\mathrm{Hom}(E'_{\mathrm{coh}}, \mathbf{Set})$ consisting of left exact functors. Hence giving a sheaf φ as in 4.13.2 is equivalent to giving a functor

$$F : {E''}^0_{\mathrm{coh}} \times E'_{\mathrm{coh}} \to \mathbf{Set} \tag{4.13.3}$$

which is a sheaf in the first variable and left exact in the second; equivalently, to giving a left exact functor

$$f_0 : E'_{\mathrm{coh}} \to E''. \tag{4.13.4}$$

If f is the associated gluing functor satisfying 4.10(iv), then f_0 is canonically isomorphic to the restriction of f to E'_{coh}. Thus, under the hypotheses considered, f_0 determines f, up to canonical isomorphism.

Corollary 4.14. When E' is a perfect topos, the category of gluing functors $f : E' \to E''$ which give a coherent glued topos E is equivalent, via $f \mapsto f_0 = f|E'_{\mathrm{coh}}$, to the category of left exact functors $f_0 : E'_{\mathrm{coh}} \to E''$, or equivalently to the category of sheaves on the site E''_{coh} (equivalently on the topos E'') with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$.

The first assertion also follows directly from 4.10, since $E' \simeq \mathrm{Ind}(E'_{\mathrm{coh}})$: for any category P with small filtered inductive limits, functors $E' \to P$ commuting with filtered inductive limits are equivalent, by restriction, to arbitrary functors $E'_{\mathrm{coh}} \to P$. If P has finite inductive limits, such a functor is left exact precisely when its restriction is.

Remarks 4.15.

a) The proof of 4.10 shows that conditions equivalent to (ii ter) and (iii ter) are obtained by replacing $\mathrm{Point}(E'')$ by any subcategory which defines a conservative family of fibre functors.

b) Taking E'' to be the punctual topos, one sees immediately that (ii ter) and (iii ter) are equivalent only when $E'_{\mathrm{coh}} = E'_{PF}$, that is, when E' is perfect. Thus the condition "E coherent" is equivalent to "f commutes with filtered inductive limits" only when E' is perfect. On the other hand, more generally taking E'' coherent of the form $\widehat{C}$, one sees that in this case condition (i) is equivalent to (iv) for every E'. In fact, in the general case the writer has not managed to construct an example where E is coherent but condition (iv) fails. Shameful!

5. Commutation of the functors $H^i(X,-)$ with filtered inductive limits

Theorem 5.1. Let E' be an algebraic topos, E a locally coherent topos, and $f : E' \to E$ a coherent morphism of topoi. Then, for every integer q, the functors $R^q f_*$ commute with filtered inductive limits of abelian sheaves.

Corollary 5.2. Let E' be a coherent topos. For every integer q, the functor $H^q(E',-)$ commutes with filtered inductive limits of abelian sheaves.

Corollary 5.3. Let E be a topos (respectively an algebraic topos), and let X be an algebraic and coherent (respectively coherent) object of E. For every integer q, the functor $H^q(X,-)$ commutes with filtered inductive limits of abelian sheaves.

Corollary 5.2 follows from 5.1 by taking E to be the punctual topos and $f : E' \to E$ the unique morphism. Conversely, 5.2 implies 5.3 by applying it to the localization morphism $j_X : E_{/X} \to E$, using the canonical isomorphism

$$H^q(X,F) \simeq H^q(E_{/X}, j_X^* F),$$

the fact that j_X^* commutes with inductive limits, and the coherence of $E_{/X}$. Finally, 5.3 implies 5.1: if $C = E_{\mathrm{coh.alg}}$ is the generating subcategory of algebraic coherent objects, then $R^q f_* F$ is the sheaf associated with the presheaf

$$X \longmapsto H^q(f^* X, F), \qquad X \in C.$$

Since f is coherent, f^*X is coherent for every $X \in C$; by 5.3 the presheaf above commutes with filtered inductive limits in F, and sheafification also commutes with inductive limits.

It remains to prove 5.2. We already know that $H^0(E',-)$ commutes with filtered inductive limits. By a classical result it is enough to prove that a filtered inductive limit of sheaves acyclic for $H^0(E',-)$ is again acyclic. This follows from the following lemma, applied to $C = E'_{\mathrm{coh}}$.

Lemma 5.4. Let E be a locally coherent topos and let C be a full generating subcategory of E, stable under fibre products, such that every object of C is quasi-compact. A filtered inductive limit of C-acyclic sheaves is C-acyclic.

Let $(F_i)_{i \in I}$ be a filtered inductive system of C-acyclic sheaves, and let $F = \varinjlim F_i$. Every $Y \in C$ is coherent, hence $F(Y) = \varinjlim_i F_i(Y)$. To prove F C-acyclic it is enough to show $\check{H}^q(Y,F) = 0$ for $q > 0$ and $Y \in C$. For a finite covering family $\mathfrak{X} = (X_\alpha \to X)$ by objects of C, the associated Cech complex uses only finite products of groups, and therefore

$$C^\bullet(\mathfrak{X}, F) = \varinjlim_i C^\bullet(\mathfrak{X}, F_i).$$

Filtered colimits are exact in abelian groups, so $H^q(\mathfrak{X},F) = \varinjlim_i H^q(\mathfrak{X}, F_i) = 0$ for $q > 0$. Passing to the limit over such coverings gives $\check{H}^q(X,F) = 0$.

Corollary 5.5. Let E be a coherent topos, and let $(X_i)_{i \in I}$ be a filtered decreasing family of coherent subobjects of the final object. Let Φ be the family of closed subtopoi contained in the closed complement of one of the X_i. Then Φ is a family of supports of E, and for every integer q the functors $H^q_\Phi(E,-)$ and $\mathcal{H}^q_\Phi$ commute with filtered inductive limits.

Let Z_i be the closed complement of X_i. For every abelian sheaf F one has the exact sequence

$$0 \to H^0_{Z_i}(E,F) \to H^0(E,F) \to H^0(X_i,F) \to H^1_{Z_i}(E,F) \to \cdots .$$

By 5.2 and 5.3 the functors $H^q(E,F)$ and $H^q(X_i,F)$ commute with filtered inductive limits in F. Hence so does $H^q_{Z_i}(E,F)$, and passing to the inductive limit over the Z_i gives the same assertion for $H^q_\Phi(E,F)$. The assertion for $\mathcal{H}^q_\Phi$ follows by localization and sheafification.

Definition 5.6. Let (E,A) be a ringed topos and let q be an integer. An A-module G is said to be of *finite q-presentation* if there is a family (X_i) of objects covering the final object of E such that, on each $E_{/X_i}$, there is an exact sequence of modules

$$L_q \to L_{q-1} \to \cdots \to L_1 \to L_0 \to G_{/X_i} \to 0, \tag{5.5.1}$$

where $L_p = A^{n_p}_{/X_i}$ for integers n_p.

Proposition 5.7. If G is a sheaf of finite q-presentation, then for every $i \leq q-1$, the functor $\mathcal{E}xt^i_A(G,-)$ commutes with filtered inductive limits of A-modules.

The question is local on E. We may therefore assume an exact sequence as in 5.5.1. Put $L_\bullet = L_q \to \cdots \to L_0$. Then

$$\mathcal{E}xt^i_A(G,F) = \mathcal{H}^i(\mathcal{H}om_A(L_\bullet,F)),$$

and $F \mapsto \mathcal{H}om_A(L_\bullet,F)$ commutes with inductive limits.

Corollary 5.8. Let (E,A) be a ringed topos, let G be a sheaf of A-modules of finite q-presentation, and let X be an algebraic and coherent object of E. The functors

$$F \longmapsto \operatorname{Ext}^i_A(X;G,F)$$

for $\frac{i(i-1)}{2} \leq q-1$ commute with filtered inductive limits in F.

This follows from 5.7 and 5.3 by the spectral sequence V 6.1.3.

5.9

Let (E,A) be a ringed topos, Φ a family of supports of E, G an A-module of finite r-presentation, and X an algebraic coherent object of E. Passing to the inductive limit over the closed subtopoi in Φ in the last spectral sequences of V 6.9.1 and V 6.9.2 gives spectral sequences

$$\begin{cases} {}'E_2^{pq} = \varinjlim_{Z\in\Phi} \operatorname{Ext}^p_A(X;G,H^q_Z F) \Rightarrow \operatorname{Ext}^{p+q}_{A,\Phi}(X;G,F), \\ {}''E_2^{pq} = \varinjlim_{Z\in\Phi} \mathcal{E}xt^p_A(G,H^q_Z F) \Rightarrow \mathcal{E}xt^{p+q}_{A,\Phi}(G,F). \end{cases} \tag{5.9.1}$$

By 5.7 and 5.8 there are canonical isomorphisms

$$\begin{cases} {}'E_2^{pq} \simeq \operatorname{Ext}^p_A(X;G,H^q_\Phi F), & \frac{p(p-1)}{2} \leq r-1, \\ {}''E_2^{pq} \simeq \mathcal{E}xt^p_A(G,H^q_\Phi F), & p \leq r-1. \end{cases} \tag{5.9.2}$$

Corollary 5.10. Use the hypotheses and notation of 5.5. Let A be a ring of E and G an A-module of finite q-presentation. For every integer i such that $\frac{i(i-1)}{2} \leq r-1$, the functors

$$F \mapsto \operatorname{Ext}^i_{A,\Phi}(E;G,F), \qquad F \mapsto \mathcal{E}xt^i_{A,\Phi}(G,F)$$

commute with filtered inductive limits.

This follows from 5.5 and 5.7 by the first spectral sequences of V 6.9.1 and V 6.9.2.

6. Inductive and projective limits of a fibred category

6.0

We assume the reader familiar with the theory of fibred categories. This section is mainly intended to fix terminology and notation. Unless otherwise stated, all categories considered here belong to a fixed universe $\mathcal{U}$.

Let $\mathcal{F}$, $\mathcal{G}$, and $\mathcal{E}$ be categories, and let $\pi : \mathcal{F} \to \mathcal{E}$, $\pi' : \mathcal{G} \to \mathcal{E}$ be functors. We write $\mathcal{H}om_{\mathcal{E}}(\mathcal{F}, \mathcal{G})$ for the category whose objects are functors $u : \mathcal{F} \to \mathcal{G}$ over $\mathcal{E}$, that is, with $\pi' u = \pi$, and whose morphisms are $\mathcal{E}$-morphisms of functors, namely natural transformations sent by π' to identity morphisms.

For an object ξ of $\mathcal{E}$, the objects X of $\mathcal{F}$ with $\pi(X) = \xi$ are called the objects of $\mathcal{F}$ over ξ. If $f : \xi \to \eta$ is a morphism of $\mathcal{E}$, morphisms in $\mathcal{F}$ over f are those mapped to f. For X over ξ and Y over η, $\mathrm{Hom}_f(X, Y)$ denotes the set of morphisms $X \to Y$ over f.

Definition 6.1.

1) A morphism $m : X \to Y$ of $\mathcal{F}$ is *cartesian* if, for every morphism $p : Z \to Y$ over $\pi(m)$, there exists a unique morphism $q : Z \to X$ over $\mathrm{id}_{\pi(X)}$ such that $mq = p$.

2) The category $\mathcal{F}$ is *prefibred* by π over $\mathcal{E}$ if for every morphism $f : \xi \to \eta$ of $\mathcal{E}$, every object of $\mathcal{F}$ over η is the target of a cartesian morphism over f.

3) The category $\mathcal{F}$ is *fibred* by π over $\mathcal{E}$ if it is prefibred and if the composite of two composable cartesian morphisms of $\mathcal{F}$ is cartesian.

6.1.1

For $\xi \in \mathcal{E}$, the *fibre category* of $\mathcal{F}$ at ξ, denoted $\mathcal{F}_\xi$, is the subcategory whose objects lie over ξ and whose morphisms lie over id_ξ. For $X, Y \in \mathcal{F}_\xi$, write $\mathrm{Hom}_\xi(X, Y)$ for the morphisms in this fibre.

6.1.2

Let $f : \xi \to \eta$ be a morphism of $\mathcal{E}$. An object Y over η is the target of a cartesian morphism over f if and only if the functor

$$Z \longmapsto \mathrm{Hom}_f(Z, Y), \qquad Z \in \mathrm{ob}(\mathcal{F}_\xi),$$

is representable. When $\mathcal{F}$ is prefibred, this gives, up to unique isomorphism, a functor $f^* : \mathcal{F}_\eta \to \mathcal{F}_\xi$, called inverse image for f. After choosing inverse-image functors for all arrows of $\mathcal{E}$, $\mathcal{F}$ is fibred if and only if $f^* g^*$ is again an inverse-image functor for every composable pair f, g. There is then a canonical isomorphism

$$C_{f,g} : f^* g^* \to (gf)^* \tag{6.1.2.1}$$

satisfying the cocycle relation

$$C_{f,hg}(f^* C_{g,h}) = C_{gh,f}(C_{f,g} h^*). \tag{6.1.2.2}$$

6.1.3

Conversely, given categories $\mathcal{F}_\xi$ for each $\xi \in \mathcal{E}$, functors $f^* : \mathcal{F}_\eta \to \mathcal{F}_\xi$ for each $f : \xi \to \eta$, and isomorphisms $C_{f,g} : f^* g^* \to (fg)^*$ satisfying the cocycle condition, one constructs, in an essentially unique way, a category $\mathcal{F}$ fibred over $\mathcal{E}$ with these fibres and inverse-image functors.

6.1.4

For two categories $\mathcal{F}$, $\mathcal{G}$ over $\mathcal{E}$, write

$$\mathcal{H}om_{\mathrm{cart}/\mathcal{E}}(\mathcal{F},\mathcal{G})$$

for the full subcategory of $\mathcal{H}om_{\mathcal{E}}(\mathcal{F},\mathcal{G})$ consisting of functors which send cartesian morphisms of $\mathcal{F}$ to cartesian morphisms of $\mathcal{G}$. Its objects are called *cartesian functors*.

Let C, C' be categories and S a set of morphisms of C. Write $\mathrm{Hom}_{S^{-1}}(C,C')$ for the set of functors $C \to C'$ sending every morphism of S to an isomorphism. Let $(\mathbf{Cat})$ be the category of categories belonging to the universe, with functors as morphisms.

Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let S be its set of cartesian morphisms. The category $\mathcal{F}$ defines two set-valued functors on $(\mathbf{Cat})$:

$$\begin{aligned} C &\longmapsto \mathrm{Hom}_{S^{-1}}(\mathcal{F},C),\\ C &\longmapsto \mathrm{Hom}_{\mathrm{Cart}\,/\mathcal{E}}(\mathcal{F},C\times\mathcal{E}),\end{aligned}$$

where $C\times\mathcal{E}$ is viewed over $\mathcal{E}$ by the second projection.

Proposition 6.2. These two functors $(\mathbf{Cat}) \to (\mathbf{Set})$ are canonically isomorphic and are representable.

For the first assertion, note that the cartesian morphisms of $C\times\mathcal{E}$ are exactly the morphisms $m\times f$ with m an isomorphism of C. For representability, one formally adjoins inverses to the morphisms of S, constructing the free category on the objects and morphisms of $\mathcal{F}$ and formal inverses of arrows of S, then quotienting by the relations of $\mathcal{F}$ and by $s^{-1}s = \mathrm{id}$, $ss^{-1} = \mathrm{id}$. The resulting category is denoted $\mathcal{F}(S^{-1})$; the canonical functor $Q : \mathcal{F} \to \mathcal{F}(S^{-1})$ represents $\mathrm{Hom}_{S^{-1}}(\mathcal{F},-)$.

Definition 6.3. When $\mathcal{F}$ is fibred over $\mathcal{E}$, the functor $\mathrm{Hom}_{S^{-1}}(\mathcal{F},-)$ is denoted $\varinjlim_{\mathcal{E}^0}\mathcal{F}$ and called the inductive-limit functor of $\mathcal{F}$ over $\mathcal{E}^0$. The representing category is also denoted $\varinjlim_{\mathcal{E}^0}\mathcal{F}$ and called the inductive-limit category of $\mathcal{F}$ over $\mathcal{E}^0$.

6.4.0

Assume $\mathcal{F}$ fibred over $\mathcal{E}$, choose inverse-image functors for all arrows of $\mathcal{E}$, and let

$$Q : \mathcal{F} \to \varinjlim_{\mathcal{E}^0}\mathcal{F}$$

be the canonical functor. The inclusions of the fibres, composed with Q, give for every $\xi\in\mathcal{E}$ a functor

$$u_\xi : \mathcal{F}_\xi \to \varinjlim_{\mathcal{E}^0}\mathcal{F},$$

and for every arrow $f : \xi \to \eta$ a diagram commuting up to canonical isomorphism

$$\begin{tikzcd} \mathcal{F}_\eta \arrow[dr, "u_\eta"] \arrow[dd, "f^*"'] & \\ & \varinjlim_{\mathcal{E}^0}\mathcal{F} \\ \mathcal{F}_\xi \arrow[ur, "u_\xi"'] & \end{tikzcd}.$$

Thus $\varinjlim_{\mathcal{E}^0} \mathcal{F}$ is the inductive limit, in the sense of pseudofunctors, of the pseudofunctor $\mathcal{E}^0 \to \mathbf{Cat}$ sending ξ to $\mathcal{F}_\xi$ and $f : \xi \to \eta$ to f^*. Even if $\xi \mapsto \mathcal{F}_\xi$ is an actual functor, this category is not generally the ordinary inductive limit in the sense of I 2.

Proposition 6.4. Let $\mathcal{F}$ be a fibred category over $\mathcal{E}$. Assume that $\mathcal{E}$ has the following properties:

L1) every span can be embedded in a commutative square;

L2) if $u, v : \bullet \rightrightarrows \bullet$ become equal after postcomposition with some morphism t, then they become equal after precomposition with some morphism w.

For example, if $\mathcal{E}^0$ is pseudo-filtered, these properties hold. Then the set S of cartesian morphisms of $\mathcal{F}$ has the following properties:

Fr1) S is stable under composition, and all isomorphisms lie in S.

Fr2) every diagram $A \to B \leftarrow C$, with the arrow $C \to B$ in S, can be completed to a commutative square whose new arrow to A is in S.

Fr3) if u, v become equal after postcomposition with an arrow of S, then they become equal after precomposition with an arrow of S.

The proof is left to the reader.

Proposition 6.5. Let $\mathcal{F}$ be a category and S a set of morphisms satisfying Fr1–Fr3. For every object X of $\mathcal{F}$, let $S(X)$ be the category of arrows in S with target X. Then $S(X)$ is cofiltered, and the localization $\mathcal{F}(S^{-1})$ can be described as follows:

a) The objects are the objects of $\mathcal{F}$.

b) For objects X, Y,

$$\mathrm{Hom}_{\mathcal{F}(S^{-1})}(X, Y) = \varinjlim_{S(X)} \mathrm{Hom}_{\mathcal{F}}(\,\cdot\,, Y).$$

c) Composition is represented by the usual calculus of right fractions: choose representatives $t : Z \to X$ in S and $f' : Z \to Y$ for f, and $t' : W \to Y$ in S, $g' : W \to Z'$ for g; use Fr2 to dominate the common target, and represent gf by the resulting composite with denominator in S.

d) If $Q : \mathcal{F} \to \mathcal{F}(S^{-1})$ is the canonical functor, then

$$Q^* : \widehat{\mathcal{F}(S^{-1})} \to \widehat{\mathcal{F}},$$

$F \mapsto FQ$, is fully faithful and injective on objects.

e) Its left adjoint

$$Q_! : \widehat{\mathcal{F}} \to \widehat{\mathcal{F}(S^{-1})}$$

is left exact. Thus Q itself is left exact when finite projective limits are representable in $\mathcal{F}$.

The proof is referred to [1].

Exercise 6.6. Let $\mathcal{F}$ be a category and S a set of morphisms satisfying Fr1–Fr3. For every object X, let $S(X)$ be the category of morphisms in S with target X.

a) Prove that $S(X)$ is cofiltered.

b) Let $J_S(X)$ be the set of sieves on X containing a sieve generated by a morphism of S. Show that the $J_S(X)$ define a topology T on $\mathcal{F}$.

c) For T, the morphisms of S are bi-covering.

d) If $\mathcal{J}^2(X)$ is the category of bi-covering morphisms with target X, then $S(X)$ is cofinal in $\mathcal{J}^2(X)$.

e) For a presheaf G on $\mathcal{F}$, if a denotes associated sheaf for T, then

$$aG(X) \simeq \varinjlim_{S(X)} G(\,\cdot\,).$$

A presheaf is a sheaf if and only if it sends all morphisms of S to isomorphisms.

f) Choose sheafification so that its restriction to $\mathcal{F}$ is injective on objects. Let $\underline{\mathcal{F}}$ be the full subcategory of sheaves associated to representable presheaves. Then $\underline{\mathcal{F}}$ is isomorphic to $\mathcal{F}(S^{-1})$. Its objects generate the topos $\mathcal{F}^{\sim}$, and the induced topology on $\underline{\mathcal{F}}$ is the chaotic topology.

g) The canonical functor $Q : \mathcal{F} \to \underline{\mathcal{F}}$ is a morphism from the site $\underline{\mathcal{F}}$ with chaotic topology to $\mathcal{F}$ with the topology T, and induces an isomorphism of the associated topoi.

h) A functor $u : \mathcal{F} \to C$ is continuous if and only if it sends all morphisms of S to isomorphisms.

i) Such a functor factors uniquely through Q:

$$\begin{tikzcd} \mathcal{F} \arrow[rr, "u"] \arrow[dr, "Q"'] & & C \\ & \underline{\mathcal{F}} \arrow[ur, "v"'] & \end{tikzcd}.$$

j) Let $\mathrm{Pro}_S\,\mathcal{F}$ be the full subcategory of $\mathrm{Pro}\,\mathcal{F}$ consisting of pro-objects whose transition morphisms lie in S. Show that Q factors as

$$\mathcal{F} \hookrightarrow \mathrm{Pro}_S\,\mathcal{F} \to \mathcal{F}_{S^{-1}},$$

and that the second functor has a fully faithful left adjoint $A : \mathcal{F}_{S^{-1}} \to \mathrm{Pro}_S\,\mathcal{F}$. For $X \in \mathcal{F}$, $AQ(X)$ is the pro-object $(Y \to X \in S(X)) \mapsto Y$.

Proposition 6.7. Let $\mathcal{E}$ be cofiltered and $\mathcal{F} \to \mathcal{E}$ a fibred category.

1) For $\xi \in \mathcal{E}$, let $\mathcal{F}/\xi$ be the fibred category over $\mathcal{E}/\xi$ obtained by base change. The canonical functor

$$\varinjlim_{\mathcal{E}/\xi} \mathcal{F}/\xi \to \varinjlim_{\mathcal{E}} \mathcal{F}$$

is an equivalence.

2) More generally, if $\mathcal{E}' \to \mathcal{E}$ is cofinal and $\mathcal{F}' \to \mathcal{E}'$ is obtained from $\mathcal{F}$ by base change, then

$$\varinjlim_{\mathcal{E}'} \mathcal{F}' \to \varinjlim_{\mathcal{E}} \mathcal{F}$$

is an equivalence of categories.

Proof: exercise.

Exercise 6.8.

1) Let $g : \mathcal{E}^0 \to \mathbf{Cat}$ be a functor and $\mathcal{G} \to \mathcal{E}$ the corresponding fibred category. Show that there is a canonical functor $\varinjlim_{\mathcal{E}^0} \mathcal{G} \to \varinjlim_{\mathcal{E}^0} g$. Show that it need not be an equivalence; one may take g constant with value the final object of $\mathbf{Cat}$ and $\mathcal{E}$ the category associated with a group.

2) If $\mathcal{E}^0$ is pseudo-filtered, show that the canonical functor above is an equivalence.

Proposition 6.9. Let $\mathcal{F} \to \mathcal{E}$ be a fibred category. The functor $(\mathbf{Cat}) \to (\mathbf{Set})$

$$C \longmapsto \mathrm{Hom}_{\mathrm{Cart}\,/\mathcal{E}}(C \times \mathcal{E}, \mathcal{F})$$

is represented by the category $\mathcal{H}om_{\mathrm{Cart}\,/\mathcal{E}}(\mathcal{E}, \mathcal{F})$.

Indeed, if $F : C \times \mathcal{E} \to \mathcal{F}$ is a cartesian $\mathcal{E}$-functor, then for every $X \in C$ the functor $\xi \mapsto F(X, \xi)$ is a cartesian functor from $\mathcal{E}$ to $\mathcal{F}$, depending functorially on X. This identifies $\mathrm{Hom}_{\mathrm{Cart}\,/\mathcal{E}}(C \times \mathcal{E}, \mathcal{F})$ with $\mathrm{Hom}(C, \mathcal{H}om_{\mathrm{Cart}\,/\mathcal{E}}(\mathcal{E}, \mathcal{F}))$.

6.10

The category $\mathcal{H}om_{\mathrm{Cart}\,/\mathcal{E}}(\mathcal{E}, \mathcal{F})$ is called the category of *cartesian sections* of the fibred category $\mathcal{F}$. It is sometimes also called the projective-limit category of $\mathcal{F}$ over $\mathcal{E}^0$, and denoted $\varprojlim_{\mathcal{E}^0} \mathcal{F}$.

6.11

Choose a cleavage of $\mathcal{F} \to \mathcal{E}$, that is, for every arrow $f : \xi \to \eta$ in $\mathcal{E}$ choose a base-change functor $f^* : \mathcal{F}_\eta \to \mathcal{F}_\xi$. For $\xi \in \mathcal{E}$, let

$$v_\xi : \varprojlim_{\mathcal{E}^0} \mathcal{F} \to \mathcal{F}_\xi$$

be evaluation at ξ. For every $f : \xi \to \eta$ there is, up to canonical isomorphism, a commutative diagram

$$\begin{tikzcd} \varprojlim_{\mathcal{E}^0} \mathcal{F} \arrow[r, "v_\xi"] \arrow[dr, "v_\eta"'] & \mathcal{F}_\xi \arrow[d, "f^*"] \\ & \mathcal{F}_\eta . \end{tikzcd}$$

Thus $\varprojlim_{\mathcal{E}^0} \mathcal{F}$ is the projective limit, in the sense of pseudofunctors, of $\xi \mapsto \mathcal{F}_\xi$. Even when this is an actual functor, this pseudofunctorial projective limit is not generally equivalent to the ordinary projective limit in the sense of I 2.

7. Fibred topoi and sites

7.1. Fibred topoi

Definition 7.1.1. Let $\mathcal{U}$ be a universe. A $\mathcal{U}$-topos fibred over a category I is a fibred category

$$p : F \to I$$

whose fibres are $\mathcal{U}$-topoi, and whose inverse-image functors relative to morphisms $f : i \to j$ of I are inverse-image functors $f^* : F_j \to F_i$ of morphisms of topoi $F_i \to F_j$.

7.1.2

Equivalently, a $\mathcal{U}$-topos fibred over I is a fibred category $p : F \to I$ whose fibres are $\mathcal{U}$-topoi and whose inverse-image functors are exact and commute with inductive limits.

7.1.3

Choose, for every arrow $f : i \to j$ of I, an inverse-image functor $f^* : F_j \to F_i$. For every composable pair $i \xrightarrow{f} j \xrightarrow{g} k$ there is a canonical isomorphism $c_{f,g} : f^*g^* \to (gf)^*$ satisfying the usual cocycle condition. Choose also for every $f : i \to j$ a right adjoint $f_* : F_i \to F_j$ to f^*. By adjunction one obtains canonical isomorphisms

$$c'_{g,f} : g_*f_* \to (gf)_* \tag{7.1.3.1}$$

satisfying the analogous cocycle condition. Thus one constructs a fibred category over I^0,

$$p' : F' \to I^0, \tag{7.1.3.2}$$

well-defined up to unique I^0-isomorphism. Its fibre over i is identified with F_i, and its inverse-image functors are the direct-image functors of the original morphisms of topoi.

7.1.4

Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over I. Choosing, for each $f : i \to j$, a morphism of topoi $F_i \to F_j$ whose inverse image is the fibred inverse-image functor, and choosing the compatible cocycles, is called a *bisplitting* of the fibred topos. Such a choice amounts to a pseudofunctor $I \to (\mathcal{U}\text{-topoi})$. Conversely every such pseudofunctor gives, essentially uniquely, a $\mathcal{U}$-topos fibred over I with bisplitting. In practice a fibred topos will often be denoted $(F_i)_{i \in I}$, with the bisplitting implicit; the constructions below do not depend on this auxiliary choice.

Definition 7.1.5. Let $p : F \to I$ and $q : G \to I$ be $\mathcal{U}$-topoi fibred over I. A morphism $m : F \to G$ of fibred topoi consists of an I-functor $m^* : G \to F$ and, for every object i of I, a right adjoint $m_{i*} : F_i \to G_i$ to the induced functor $m_i^* : G_i \to F_i$. Each adjoint pair (m_i^*, m_{i*}) is required to be a morphism of topoi $F_i \to G_i$.

7.1.6

Let $m : F \to G$ be a morphism of $\mathcal{U}$-topoi fibred over I. The construction 7.1.3 associates fibred categories $F' \to I^0$, $G' \to I^0$. Choosing bisplittings, the I-functor m^* gives transition morphisms

$$b_f : f^* m_j^* \to m_i^* f^* \tag{7.1.6.1}$$

compatible with cocycles:

$$m_i^*(c_{f,g})\, b_f^*\, f^*(b_g) = b_{gf} c_{f,g}. \tag{7.1.6.2}$$

Passing to adjoints gives morphisms

$$b'_f : f_* m_{i*} \to m_{j*} f_* \tag{7.1.6.3}$$

with analogous relations. Hence one constructs an I^0-functor

$$m_* : F' \to G' \tag{7.1.6.4}$$

whose restrictions to fibres are the m_{i*}. The resulting functor is independent, up to canonical isomorphism, of the auxiliary choices.

7.1.7

Let $\mathrm{Hom}_{\mathrm{top},I}(F,G)$ be the set of morphisms of fibred topoi from F to G. It is the set of objects of a category $\mathcal{H}om_{\mathrm{top},I}(F,G)$: a morphism $m \to n$ is an I-morphism of functors $n^* \to m^*$, equivalently, by adjunction, an I^0-morphism $m_* \to n_*$. Thus one has two functors

$$\mathcal{H}om_{\mathrm{top},I}(F,G) \to \mathcal{H}om_I(G,F)^0, \tag{7.1.7.1}$$

$$\mathcal{H}om_{\mathrm{top},I}(F,G) \to \mathcal{H}om_{I^0}(F',G'). \tag{7.1.7.2}$$

They send a morphism $(m^*, (m_{i*}))$ respectively to its inverse-image functor $m^* : G \to F$, and to its direct-image functor $m_* : F' \to G'$. These functors are fully faithful. Their essential images are, respectively, the I-functors $G \to F$ which fibrewise are exact and commute with inductive limits, and the I^0-functors $F' \to G'$ which fibrewise have an exact left adjoint. This justifies the common abuse of specifying a morphism of fibred topoi by either its inverse or its direct image; the morphism is then defined only up to unique isomorphism.

A morphism is called *cartesian* if m_*, equivalently m^*, is a cartesian functor. The corresponding full subcategory $\mathcal{H}om_{\mathrm{top},\mathrm{Cart}\,/I}(F,G)$ embeds fully faithfully as

$$\mathcal{H}om_{\mathrm{top},\mathrm{Cart}\,/I}(F,G) \to \mathcal{H}om_{\mathrm{cart}/I}(G,F)^0, \tag{7.1.7.3}$$

$$\mathcal{H}om_{\mathrm{top},\mathrm{Cart}\,/I}(F,G) \to \mathcal{H}om_{\mathrm{cart}/I^0}(F',G'). \tag{7.1.7.4}$$

7.1.8

Morphisms of $\mathcal{U}$-topoi fibred over I compose as ordinary morphisms of topoi do. For a universe $\mathcal{V}$ with $\mathcal{U} \in \mathcal{V}$, one obtains a category $(\mathcal{V}\text{-}\mathcal{U}\text{-}\,\mathrm{Top}_{/I})$ whose objects are $\mathcal{U}$-topoi fibred over I with fibre categories in $\mathcal{V}$, and whose morphisms are morphisms of $\mathcal{U}$-topoi fibred over I. The usual functoriality of composition and the considerations of IV 3.3 and IV 3.4 extend without change. When I is the punctual category, this reduces to the usual theory of morphisms of $\mathcal{U}$-topoi.

7.1.9

Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over I, and let $\phi : J \to I$ be a functor. The base-changed fibred category $F_J \to J$ is again a $\mathcal{U}$-topos fibred over J. The construction $F \mapsto F'$ of 7.1.3 is compatible with this base change. Base change is functorial, indeed 2-functorial, and for fibred topoi F, G over I one has commutative diagrams

$$\begin{tikzcd} \mathcal{H}om_{\mathrm{top},I}(F,G) \arrow[r] \arrow[d] & \mathcal{H}om_I(G,F)^0 \arrow[d] \\ \mathcal{H}om_{\mathrm{top},J}(F_J,G_J) \arrow[r] & \mathcal{H}om_J(G_J,F_J)^0, \end{tikzcd} \tag{7.1.9.1}$$

and

$$\begin{tikzcd} \mathcal{H}om_{\mathrm{top},I}(F,G) \arrow[r] \arrow[d] & \mathcal{H}om_{I^0}(F',G') \arrow[d] \\ \mathcal{H}om_{\mathrm{top},J}(F_J,G_J) \arrow[r] & \mathcal{H}om_{J^0}(F'_{J^0},G'_{J^0}), \end{tikzcd} \tag{7.1.9.2}$$

where the vertical functors are base change and the horizontal functors are respectively inverse image and direct image.

7.2. Fibred sites

7.2.1

A *$\mathcal{U}$-fibred site* C over a category I is a fibred category $p : C \to I$ whose fibres are endowed with topologies making them $\mathcal{U}$-sites, such that for every arrow $f : i \to j$ of I, the inverse-image functor $f^* : C_j \to C_i$ is a morphism of sites from C_i to C_j.

7.2.2

Let $p : C \to I$, $q : D \to I$ be $\mathcal{U}$-fibred sites. A *morphism of $\mathcal{U}$-fibred sites from C to D* is an I-functor $m : D \to C$ such that, for every object i of I, the induced functor $m_i : D_i \to C_i$ is a morphism of sites from C_i to D_i. Write $\mathrm{MorSite}_I(C, D)$ for the set of such morphisms. A *continuous functor from D to C* is an I-functor inducing continuous functors on fibres. One obtains fully faithful inclusions

$$\mathcal{M}\mathrm{orSite}_I(C,D) \hookrightarrow \mathcal{C}\mathrm{ont}_I(D,C)^0 \hookrightarrow \mathcal{H}om_I(D,C)^0. \tag{7.2.2.1}$$

Cartesian morphisms are defined as in 7.1.7, giving a diagram of full subcategories

$$\begin{tikzcd} \mathcal{M}\mathrm{orSite}_{\mathrm{Cart}\,/I}(C,D) \arrow[r, hook] \arrow[d, hook] & \mathcal{H}om_{\mathrm{Cart}\,/I}(D,C)^0 \arrow[d, hook] \\ \mathcal{M}\mathrm{orSite}_I(C,D) \arrow[r, hook] & \mathcal{H}om_I(D,C)^0. \end{tikzcd} \tag{7.2.2.2}$$

7.2.3

Morphisms of $\mathcal{U}$-fibred sites compose. Thus one defines the category $(\mathcal{V}\text{-}\mathcal{U}\text{-Site}/I)$, whose objects are $\mathcal{U}$-sites with fibres belonging to a universe $\mathcal{V}$ containing $\mathcal{U}$. This category has a natural 2-categorical structure, and the composition of morphisms is functorial with respect to morphisms between morphisms. For every $\phi : J \to I$ and every fibred site $C \to I$, the base change $C_J \to J$ is canonically a $\mathcal{U}$-fibred site; base change is 2-functorial.

7.2.4

Let $p : C \to I$ be a fibred category whose fibres are $\mathcal{U}$-sites, and assume finite projective limits are representable in the fibres. Then p is a $\mathcal{U}$-fibred site if the inverse-image functors are left exact and carry covering families to covering families; this condition is also necessary when the fibre topologies are no finer than the canonical topology. All fibred sites used in the Seminar will be of this type. Similarly, for fibred sites $p : C \to I$, $q : D \to I$ with finite projective limits in the fibres, a cartesian functor $m : D \to C$ whose fibre functors are left exact and carry covering families to covering families is a morphism of fibred sites from C to D; conversely under the same canonical-topology hypothesis. All morphisms of fibred sites used here will be of this type.

7.2.5

A $\mathcal{U}$-topos fibred over I is a $\mathcal{U}$-fibred site when its fibres are endowed with their canonical topologies, and this convention will be used below. If $m : F \to G$ is a morphism of fibred topoi, then its inverse-image functor $m^* : G \to F$ is a morphism of fibred sites from F to G. Hence there is a functor

$$(\mathcal{V}\text{-}\mathcal{U}\text{- Top}_{/I}) \to (\mathcal{V}\text{-}\mathcal{U}\text{-Site}_{/I}),$$

not in general faithful. It naturally extends to a 2-functor which, for two fibred topoi F, G, induces an equivalence

$$\mathcal{H}om_{\mathrm{top},I}(F, G) \to \mathcal{M}\mathrm{orSite}_I(F, G). \tag{7.2.5.1}$$

Composed with the inclusion into $\mathcal{H}om_I(G, F)^0$, this is precisely the inverse-image functor 7.1.7.1.

7.2.6

Let $p : C \to I$ be a $\mathcal{U}$-fibred site. Choose inverse-image functors $f^* : C_j \to C_i$. Passing to sheaf categories, f^* extends to a functor $\mathrm{Top}(f)^* : C_j^\sim \to C_i^\sim$, where $\mathrm{Top}(f) : C_i^\sim \to C_j^\sim$ is a morphism of topoi, giving a commutative square up to isomorphism

$$\begin{tikzcd}
C_j \arrow[r, "f^*"] \arrow[d, "\varepsilon_j"'] & C_i \arrow[d, "\varepsilon_i"] \\
C_j^\sim \arrow[r, "\mathrm{Top}(f)^*"] & C_i^\sim .
\end{tikzcd} \tag{7.2.6.1}$$

Together with the canonical cocycles, these data define a fibred category

$$p^\sim : C^{\sim/I} \to I, \tag{7.2.6.6}$$

whose fibres are the topoi $C_i^\sim$, and a cartesian functor

$$\varepsilon_{C/I} : C \to C^{\sim/I}. \tag{7.2.6.7}$$

The fibred topos $C^{\sim/I}$ is independent, up to canonical isomorphism, of all choices and is called the *$\mathcal{U}$-fibred topos associated* with the $\mathcal{U}$-fibred site C.

7.2.6.8 The formation of the associated $\mathcal{U}$-fibred topos commutes with base change of the base category.

Proposition 7.2.7. Let E be a $\mathcal{U}$-topos fibred over I, and C a $\mathcal{U}$-fibred site over I. The functor which sends a morphism of fibred topoi $m : E \to C^{\sim/I}$ to the composite $m^*\varepsilon_{C/I}$ induces an equivalence of categories, preserving cartesian objects,

$$\mathcal{H}om_{\mathrm{top},I}(E, C^{\sim/I}) \xrightarrow{\sim} \mathcal{M}\mathrm{orSite}_I(E, C).$$

If finite projective limits are representable in the fibres of C, the corresponding fully faithful functor

$$\mathcal{H}om_{\mathrm{top},I}(E, C^{\sim/I}) \to \mathcal{H}om_I(C, E)^0$$

has as essential image the I-functors $g : C \to E$ which are fibrewise left exact and carry covering families in fibres to epimorphic families in the corresponding fibres.

This is exactly IV 4.9.4 in the fibred setting; its proof generalizes immediately.

7.3. Examples

Example 7.3.1: the fibred topos of morphisms of a topos. Let E be a $\mathcal{U}$-topos, let $\mathrm{Fl}(E)$ be the category of morphisms of E, and let $p : \mathrm{Fl}(E) \to E$ send a morphism to its target. Then $(\mathrm{Fl}(E), p)$ is a $\mathcal{U}$-topos fibred over E. The fibre over X is $E_{/X}$, and the inverse image along $f : X \to Y$ is fibre product.

For every functor $\phi : I \to E$ one obtains by base change a $\mathcal{U}$-topos fibred over I, denoted

$$p_I : \mathrm{Fl}(E)_I \to I, \tag{7.3.1.1}$$

whose fibre over i is $E_{/\phi(i)}$. If $m : \phi_1 \to \phi_2$ is a morphism of functors $I \to E$, then there is a unique cartesian morphism of fibred topoi

$$\mathrm{loc}(m) : \mathrm{Fl}(E)_1 \to \mathrm{Fl}(E)_2 \tag{7.3.1.4}$$

which on fibres is the localization morphism

$$\mathrm{loc}(m(i)) : E_{/\phi_1(i)} \to E_{/\phi_2(i)}. \tag{7.3.1.3}$$

For composable morphisms m, n of such functors there are canonical isomorphisms

$$c(m, n) : \mathrm{loc}(n)\,\mathrm{loc}(m) \simeq \mathrm{loc}(nm), \tag{7.3.1.5}$$

satisfying the usual cocycle condition.

Example 7.3.2: the fibred site of morphisms of a site. Let C be a $\mathcal{U}$-site in which fibre products are representable. Let $\mathrm{Fl}(C)$ be the category of morphisms of C, and let $p : \mathrm{Fl}(C) \to C$ send a morphism to its target. With the induced topologies on the fibres, this is a $\mathcal{U}$-fibred site over C. If $\varepsilon_C : C \to C^{\sim}$ is the canonical functor, its natural extension to arrow categories gives a cartesian functor

$$\mathrm{Fl}(C) \to \mathrm{Fl}(C^{\sim})_C$$

over C, hence a morphism of fibred topoi

$$\mathrm{can} : \mathrm{Fl}(C^{\sim})_C \to \mathrm{Fl}(C)^{\sim/C}. \tag{7.3.2.3}$$

This morphism is a C-equivalence of $\mathcal{U}$-fibred topoi. After base change by any $\phi : I \to C$, one obtains an I-equivalence

$$\mathrm{can}_I : \mathrm{Fl}(C^\sim)_I \to \mathrm{Fl}(C)_I^{\sim/I}. \tag{7.3.2.5}$$

If $m : \phi_1 \to \phi_2$ is a morphism of functors $I \to C$, then the localization morphisms of fibred sites $\mathrm{loc}(m) : \mathrm{Fl}(C)_1 \to \mathrm{Fl}(C)_2$ are compatible, up to isomorphism, with their counterparts after passing to $C^\sim$.

Exercise 7.3.3: small and large fibred sites. This exercise continues IV 4.10.6. Let $\mathcal{M}$ be the full subcategory of $\mathrm{Fl}(S)$ defined by the morphisms in $\mathcal{M}$, and let $p : \mathcal{M} \to S$ send a morphism to its target.

6°) Show that $(\mathcal{M}, p)$ is a fibred site over S, and, when fibre products are representable in S, the inclusion $\mathcal{M} \to \mathrm{Fl}(S)$ is a cartesian morphism from the fibred site $\mathrm{Fl}(S) \to S$ to $\mathcal{M} \to S$.

7°) Let $\mathcal{M}^{\sim/S} \to S$ be the associated fibred topos. Show that there is a canonical morphism $g : \mathrm{Fl}(S^\sim)_{/S} \to \mathcal{M}^{\sim/S}$ of fibred topoi over S, inducing on each fibre over $X \in S$ the morphism $(\mathrm{Prol}_X, \mathrm{Res}_X)$. Although each $g_X : S^\sim_{/X} \to S(X)^\sim$ has a section, show that g need not have a section in general.

7.4. The total topology of a fibred topos or site

Definition 7.4.1. Let $p : C \to I$ be a $\mathcal{U}$-fibred site, and for $i \in \mathrm{ob}\, I$ let $\alpha_{i!} : C_i \to C$ be the inclusion of the fibre. The *total topology* on C is the least topology making all the functors $\alpha_{i!}$ continuous. The category C with this topology is the *total site*.

Proposition 7.4.2. Let $p : C \to I$ be a $\mathcal{U}$-fibred site, let T be the total topology on C, and let T_i be the topology of the fibre C_i.

1) If X lies over i, a family $(X_\beta \to X)$ is covering for T if and only if it is refined by a covering family $(Y_\gamma \to X)$ for T_i inside C_i.

2) The topology induced by T on C_i is T_i.

3) For every i, the functor $\alpha_{i!} : C_i \to C$ is cocontinuous.

Property 3 follows from 1 and III 2.1. To prove 1, let $J(X)$ be the set of sieves described in 1. These $J(X)$ satisfy the axioms of a topology T'. It is the least topology making the covering families of the fibres covering in C, hence is no finer than T. It remains to prove that each $\alpha_{i!}$ is continuous for T', equivalently that the topology induced on C_i is T_i. This reduces to the following lemma.

Lemma 7.4.2.2. With the preceding notation, let $X \in C_i$ and let $u : R \to X$ be a morphism of presheaves on C_i. The following are equivalent:

(i) $R \to X$ is bi-covering for T_i.

(ii) $\alpha_{i!}(R) \to \alpha_{i!}(X)$ is bi-covering for T'.

For (i) $\Rightarrow$ (ii), covering is clear. If two maps $m, n : Y \rightrightarrows \alpha_{i!}R$ over an object Y lying over j become equal after applying $\alpha_{i!}u$, then using the formula

$$\mathrm{Hom}_{\widehat{C}}(Y, \alpha_{i!}R) \simeq \coprod_{w \in \mathrm{Hom}(j,i)} \mathrm{Hom}_{\widehat{C_j}}(Y, w^*R),$$

they correspond to maps into f^*R equal after f^*u. Since inverse-image functors are morphisms of sites, f^*u is bi-covering, so the two maps become equal after a covering of Y. Conversely, if $\alpha_{i!}u$ is bi-covering, cocontinuity of $\alpha_{i!}$ shows $\alpha_i^*\alpha_{i!}u$ bi-covering; the cartesian square

$$\begin{tikzcd} R \arrow[r] \arrow[d, "u"'] & \alpha_i^*\alpha_{i!}R \arrow[d, "\alpha_i^*\alpha_{i!}u"] \\ X \arrow[r] & \alpha_i^*\alpha_{i!}X \end{tikzcd}$$

then implies that u is bi-covering.

Remark 7.4.3.

1) Proposition 7.4.2 can be stated and proved in a slightly more general, but apparently useless, setting: the fibres have topologies and the inverse-image functors are continuous, without requiring them to be morphisms of sites.

2) The total topology of a fibred site is also the finest topology making the functors $\alpha_{i!} : C_i \to C$ cocontinuous.

3) If I is equivalent to a $\mathcal{U}$-small category, then the total site C is a $\mathcal{U}$-site: a small topological generating family is obtained by collecting such families in the fibres. The associated topos of sheaves on the total site is denoted $\mathrm{Top}(C)$ and called the *total topos*.

4) Since $\alpha_{i!} : C_i \to C$ is cocontinuous, it gives a morphism of topoi $\alpha_i : C_i^\sim \to \mathrm{Top}(C)$. Since $\alpha_{i!}$ is also continuous, the inverse-image functor $\alpha_i^* : \mathrm{Top}(C) \to C_i^\sim$ admits a left adjoint, also denoted $\alpha_{i!} : C_i^\sim \to \mathrm{Top}(C)$, extending the inclusion $C_i \to C$.

Proposition 7.4.4. Let $p : C \to I$ be a $\mathcal{U}$-fibred site over a small category I.

1) A presheaf F on C is a sheaf for the total topology if and only if, for every $i \in I$, its restriction $F|C_i = F \circ \alpha_{i!}$ is a sheaf on C_i.

2) A functor $m : C \to D$ to a site D is continuous from the total site C to D if and only if every restriction $m|C_i : C_i \to D$ is continuous.

If F is a sheaf on the total site, its restriction to every fibre is a sheaf because $\alpha_{i!}$ is continuous. Conversely, if all restrictions are sheaves, then by the description of covering sieves in 7.4.2 the sheaf condition on each fibre implies the sheaf condition on the total site. The assertion on continuous functors follows immediately from the sheaf criterion and the definition of continuity.

7.4.5

Let $p : C \to I$ be a $\mathcal{U}$-fibred site over a category equivalent to a small category, and let $f : i \to j$ be an arrow of I. Let $f : C_i^\sim \to C_j^\sim$ also denote the morphism of topoi determined by the fibred-site structure. With the notation of 7.4.3, one has a diagram of topoi

$$\begin{tikzcd} C_i^\sim \arrow[r, "\alpha_i"] \arrow[d, "f"'] & \mathrm{Top}(C) \\ C_j^\sim \arrow[ur, "\alpha_j"'] & \end{tikzcd} \tag{7.4.5.1}$$

which is not commutative in general. We define a canonical morphism of morphisms of topoi

$$\rho_f : \alpha_i \to \alpha_j \circ f. \tag{7.4.5.2}$$

At the inverse-image level this amounts to a natural morphism

$$\beta_f^* : f^*\alpha_j^* \to \alpha_i^*, \tag{7.4.5.3}$$

or, by adjunction, to

$$\gamma_f : \alpha_j^* \to f_*\alpha_i^*. \tag{7.4.5.4}$$

For a sheaf F on the total site and $Y \in C_j$, one has $\alpha_j^* F(Y) = F(\alpha_{j!}Y)$, while $f_*\alpha_i^*F(Y) = F(\alpha_{i!}f^*Y)$. The canonical cartesian morphism

$$m_Y : \alpha_{i!}f^*Y \to \alpha_{j!}Y \tag{7.4.5.5}$$

therefore yields

$$\gamma_f(F)(Y) = F(m_Y) : \alpha_j^*F(Y) \to f_*\alpha_i^*F(Y), \tag{7.4.5.6}$$

and hence the required morphism of functors. For composable $f : i \to j$, $g : j \to k$, with $c'_{g,f} : g_*f_* \to (gf)_*$, the compatibility relation is

$$c'_{g,f} \circ g_*(\gamma_f) \circ \gamma_g = \gamma_{gf}. \tag{7.4.5.8}$$

7.4.6

Let $C^{\sim/I} \to I$ be the fibred topos associated with C, and let $(C^{\sim/I})' \to I^0$ be the fibred category of direct images. The preceding construction sends a sheaf F on $\mathrm{Top}(C)$ to the family $F_i = \alpha_i^*F \in C_i^\sim$, together with morphisms

$$\gamma_f(F) : F_j \to f_*F_i,$$

satisfying the compatibility 7.4.5.8. Thus one obtains a functor

$$\Theta_C : \mathrm{Top}(C) \to \mathcal{H}om_{I^0}(I^0, (C^{\sim/I})'). \tag{7.4.6.1}$$

Proposition 7.4.7. If $C \to I$ is a $\mathcal{U}$-fibred site and I is equivalent to a small category, then Θ_C is an equivalence of categories.

A quasi-inverse is as follows. Given a section σ of $(C^{\sim/I})'$ over I^0, define a presheaf on C by

$$X \longmapsto \sigma(p(X))(X).$$

For a morphism $m : X \to Y$ over $f : i \to j$, factor m through the cartesian morphism $f^*Y \to Y$, and use the structural morphism $\sigma(j) \to f_*\sigma(i)$ to obtain the restriction map. This presheaf is a sheaf on the total site by 7.4.4, and the construction is functorial. Direct comparison with the definition of Θ_C shows that it is a quasi-inverse.

Remark 7.4.8. It follows in particular that $\mathcal{H}om_{I^0}(I^0, (C^{\sim/I})')$ is a $\mathcal{U}$-topos. This can also be checked directly using the usual axioms for topoi and the existence of a small generating category.

7.4.9

Let $p : C \to I$, $q : D \to I$ be two $\mathcal{U}$-fibred sites over a small category, and let m be a morphism of fibred sites from C to D. Let

$$m^{\sim/I} : C^{\sim/I} \to D^{\sim/I}$$

be the corresponding morphism of associated fibred topoi, and let

$$m_*^{\sim/I} : (C^{\sim/I})' \to (D^{\sim/I})'$$

be its direct-image functor. Passing to sections gives

$$\mathrm{Hom}_{I^0}(I^0, m_*^{\sim/I}) : \mathcal{H}om_{I^0}(I^0, (C^{\sim/I})') \to \mathcal{H}om_{I^0}(I^0, (D^{\sim/I})'). \tag{7.4.9.1}$$

On the other hand m, being continuous between total sites, gives a direct-image functor $m_* : \mathrm{Top}(C) \to \mathrm{Top}(D)$. Hence there is a diagram

$$\begin{tikzcd} \mathrm{Top}(C) \arrow[r, "\Theta_C"] \arrow[d, "m_*"'] & \mathcal{H}om_{I^0}(I^0, (C^{\sim/I})') \arrow[d, "\mathrm{Hom}_{I^0}(I^0, m_*^{\sim/I})"'] \\ \mathrm{Top}(D) \arrow[r, "\Theta_D"] & \mathcal{H}om_{I^0}(I^0, (D^{\sim/I})'). \end{tikzcd} \tag{7.4.9.2}$$

Proposition 7.4.10. The diagram 7.4.9.2 is commutative up to isomorphism. If m is cartesian, then m is a morphism from the total site C to the total site D.

The commutativity is a direct verification. If m is cartesian, it is enough, using 7.4.7, to show that the left adjoint to $\mathrm{Hom}_{I^0}(I^0, m_*^{\sim/I})$ is left exact. Fibrewise, write $m_i^* : D_i^{\sim} \to C_i^{\sim}$ for the inverse image. For each $f : i \to j$, adjunction and cartesianity give a canonical morphism

$$\mathrm{can}_f : m_j^* f_* \to f_* m_i^*. \tag{7.4.10.4}$$

If τ is a section of $(D^{\sim/I})'$, put $\sigma(i) = m_i^*(\tau(i))$, and define the transition morphism of σ by composing

$$m_j^*\tau(j) \xrightarrow{m_j^*\gamma_f(\tau)} m_j^* f_*\tau(i) \xrightarrow{\mathrm{can}_f} f_* m_i^*\tau(i). \tag{7.4.10.6}$$

This defines a section of $(C^{\sim/I})'$. A formal adjoint-functor chase shows this construction is the left adjoint in question. Its left exactness follows because finite projective limits of sections are computed fibrewise and all functors used in the construction commute with finite projective limits.

Corollary 7.4.11. Let $p : C \to I$ be a $\mathcal{U}$-fibred site over a small category, let $C^{\sim/I} \to I$ be its associated fibred topos, and let $\varepsilon_I : C \to C^{\sim/I}$ be the canonical morphism of fibred sites. Then ε_I induces an equivalence

$$\mathrm{Top}(\varepsilon_I) : \mathrm{Top}(C) \to \mathrm{Top}(C^{\sim/I})$$

of the corresponding total topoi.

This follows from 7.4.9.2 and the fact that ε_I is an equivalence on associated fibred topoi. Consequently, questions about the total topos may replace fibred sites by the associated fibred topoi.

Lemma 7.4.12. Let $p : C \to I$ be a fibred site over a small category I, and let i be a final object of I. The functor $\alpha_{i!} : C_i^{\sim} \to \mathrm{Top}(C)$ is exact and fully faithful. The pairs

$$\beta_i = (\alpha_{i!}, \alpha_i^*) : \mathrm{Top}(C) \to C_i^{\sim}, \qquad \alpha_i = (\alpha_i^*, \alpha_{i*}) : C_i^{\sim} \to \mathrm{Top}(C)$$

are morphisms of topoi, and the composite $\beta_i\alpha_i : C_i^{\sim} \to C_i^{\sim}$ is isomorphic to the identity.

For every $j \in I$, let $f_j : j \to i$ be the unique morphism. Using the equivalence Θ_C, the object $\alpha_{i!}(X)$ corresponds to the section $j \mapsto f_j^* X$, while α_i^* evaluates a section at i. Hence $\alpha_i^*\alpha_{i!} \simeq \mathrm{id}$, so $\alpha_{i!}$ is fully faithful. It is left exact because the f_j^* are left exact. The remaining assertions follow.

Theorem 7.4.13.1. Let $p : C \to I$ and $q : D \to I$ be $\mathcal{U}$-fibred sites over a small category I, and let $m : D \to C$ be an I-functor. For m to be a morphism from the fibred site C to the fibred site D, it is enough that m be a morphism from the total site C to the total site D.

Assume $m : D \to C$ is a morphism of total sites. We must show that for every i, $m_i : D_i \to C_i$ is a morphism of sites from C_i to D_i. First m_i is continuous: if $R \hookrightarrow X$ is a covering sieve in C_i, then, by the commutative square with the inclusions $\alpha_{i!}$ and by continuity of m and $\alpha_{i!}$, the morphism $\alpha_{i!}m_{i!}(R) \to \alpha_{i!}m_i(X)$ is bi-covering, and Lemma 7.4.2.2 implies the desired bi-covering condition in the fibre.

It remains to reduce to the case of fibred topoi. Since the m_i are continuous, they extend to sheaf categories as functors $m_i^* : D_i^{\sim} \to C_i^{\sim}$. These assemble into a cartesian functor $m^* : D^{\sim/I} \to C^{\sim/I}$. The square

$$\begin{tikzcd} D \arrow[r, "m"] \arrow[d, "\varepsilon_I"'] & C \arrow[d, "\varepsilon_I"] \\ D^{\sim/I} \arrow[r, "m^*"] & C^{\sim/I} \end{tikzcd}$$

commutes up to isomorphism. Since the vertical functors induce equivalences on total topoi, and since m is a morphism of total sites, it is enough to prove the assertion when C and D are fibred topoi.

In that case, each m_i sends the final object of D_i to the final object of C_i. Indeed, under the equivalence Θ_D, a sheaf on the total topos is a section, and m sends a section σ to $i \mapsto m_i\sigma(i)$. Since m is left exact, it sends the final section to the final section.

For arbitrary i, replace the fibred topoi by their localizations over the final object e_i of the fibre. This reduces to the case where i is final in I. In that case, using Lemma 7.4.12, m_i identifies with the composite $\alpha_i^* m^{\sim} \alpha_{i!}$. All factors are left exact, hence m_i is left exact and therefore is a morphism of sites from C_i to D_i.

Exercise 7.4.14. Let $X = (X_i)_{i\in I}$ be a fibred topos over I, and let T be a topos. Show that the categories $\mathcal{H}om_{\mathrm{top}}(X_i, T)$ are the fibres of a fibred category denoted $\mathcal{H}om_{\mathrm{top},I}(X, T \times I)$. Show that its category of sections over I is canonically equivalent to $\mathcal{H}om_{\mathrm{top},I}(\mathrm{Top}(X),T)$, where $\mathrm{Top}(X)$ is the total topos of X.

Exercise 7.4.15. Let $X=(X_i)_{i\in I}$ be a fibred topos over I. Define a cartesian morphism m from X to the constant fibred topos with fibre (**Set**). Show that the total topos of this constant fibred topos is canonically equivalent to $\widehat{I}$, and deduce a morphism of topoi

$$\mathrm{Top}(m):\mathrm{Top}(X)\to\widehat{I}.$$

For an object $F=(F_i)_{i\in I}$ of $\mathrm{Top}(X)$, show that

$$\mathrm{Top}(m)(F)=\big(i\mapsto\Gamma(X_i,F_i)\big).$$

Deduce from 7.4.12 that if F is an injective abelian object, then each F_i is injective abelian, and hence

$$R^q\,\mathrm{Top}(m)(F)=\big(i\mapsto H^q(X_i,F_i)\big).$$

Show that the Cartan-Leray spectral sequence of m is

$$H^{p+q}(\mathrm{Top}(X),F)\Leftarrow\varprojlim_{I}{}^{(p)}H^q(X_i,F|X_i).$$

8. Projective limits of fibred topoi

8.1. Generalities

Definition 8.1.1. Let $\mathcal{U}$ be a universe and let $p:F\to I$ be a $\mathcal{U}$-topos fibred over I. A pair (C,m), consisting of a $\mathcal{U}$-topos C and a Cartesian morphism

$$m:C\times I\longrightarrow F$$

of $\mathcal{U}$-topoi fibred over I, is called a *projective limit* of the fibred topos F if, for every $\mathcal{U}$-topos D, the functor

$$\mathcal{H}om_{\mathrm{top}}(D,C)\to\mathcal{H}om_{\mathrm{top},\mathrm{Cart}\,/I}(D\times I,F)\tag{8.1.1.1}$$

obtained by base change and then by composition with m is an equivalence of categories.

8.1.2

With the notation of 8.1.1, let D be a $\mathcal{U}$-topos and let $m_1:D\times I\to F$ be a morphism of $\mathcal{U}$-topoi fibred over I. Translating the definition, one sees that there is a morphism of topoi $n:D\to C$, together with an isomorphism of fibred morphisms

$$\gamma:m_1\xrightarrow{\sim}m\circ(n\times\mathrm{id}_I).$$

If (n',γ') is another such pair, there is a unique isomorphism $\eta:n\to n'$ compatible with γ and γ'. Hence, if (D,m_1) is itself a projective limit of F, then n is an equivalence of $\mathcal{U}$-topoi; the additional commutation isomorphism determines the pair up to canonical isomorphism. Existence will be proved in 8.2.3 when I is cofiltered.

8.1.3

We write

$$\varprojlim_I F, \qquad \varprojlim_I F_i, \qquad \overleftarrow{F}$$

for a projective limit of the fibred topos F, and write

$$\mu : (\varprojlim_I F) \times I \to F$$

for the canonical morphism. For each object $i \in I$, the induced morphism on the fibre is denoted

$$\mu_i : \overleftarrow{F} \to F_i.$$

Choose a cleavage and an opcleavage for F, so that to every arrow $f : i \to j$ in I there corresponds a morphism of topoi $f. = (f^*, f_*) : F_i \to F_j$, with coherence isomorphisms $c_{f,g} : g.f. \simeq (gf).$. Since μ is Cartesian, for every $f : i \to j$ there is an isomorphism

$$b_f : f.\mu_i \xrightarrow{\sim} \mu_j, \tag{8.1.3.1}$$

and the family (b_f) satisfies the usual cocycle compatibilities.

Equivalently, if D is a $\mathcal{U}$-topos, if $m : D \times I \to F$ is a fibred morphism, if $m_i : D \to F_i$ are the fibrewise morphisms, and if

$$b'_f : f.m_i \xrightarrow{\sim} m_j$$

are the transition isomorphisms, then there is a morphism $n : D \to \overleftarrow{F}$ and isomorphisms

$$\gamma_i : m_i \xrightarrow{\sim} \mu_i n$$

compatible with the transitions. Conversely, data $\overleftarrow{F}$, μ_i, and b_f, satisfying this universal property, determine a unique fibred morphism $\mu : \overleftarrow{F} \times I \to F$, and $(\overleftarrow{F}, \mu)$ is the projective limit.

If $\mathcal{V}$ is a universe containing $\mathcal{U}$, this is precisely the 2-categorical projective limit of the pseudo-functor defined by the chosen transition morphisms $f.$. This should not be confused with a strict projective limit. Even when the pseudo-functor is an honest functor, the 2-categorical limit and the naive strict limit need not agree.

8.1.4–8.1.7

Let $p : F \to I$ and $q : G \to I$ be two $\mathcal{U}$-topoi fibred over I, and let $m : F \to G$ be a morphism of fibred topoi. If F and G have projective limits, the universal property gives a morphism

$$\overleftarrow{m} : \overleftarrow{F} \to \overleftarrow{G}, \tag{8.1.4.1}$$

together with the evident commutation isomorphism between $m \circ \mu_F$ and $\mu_G \circ (\overleftarrow{m} \times \mathrm{id}_I)$, unique up to a unique isomorphism.

For a base change $\phi : I' \to I$, write $F_\phi \to I'$ for the induced fibred topos. If $\overleftarrow{F}$ and $\overleftarrow{F_\phi}$ exist, the universal property gives a canonical morphism

$$m_\phi : \overleftarrow{F} \to \overleftarrow{F_\phi},$$

again unique up to a unique isomorphism.

Now let $p : F \to I$ be a $\mathcal{U}$-site fibred over a small category I, and let $F^{\sim/I} \to I$ be the associated fibred topos. Let D be a $\mathcal{U}$-topos. There are natural functors

$$\mathcal{H}om_{\mathrm{top,Cart}\,/I}(D \times I, F^{\sim/I}) \to \mathrm{MorSite}_{\mathrm{Cart}\,/I}(D \times I, F), \tag{8.1.6.1}$$

$$\mathrm{MorSite}_{\mathrm{Cart}\,/I}(D \times I, F) \to \mathcal{H}om_{\mathrm{Cart}\,/I}(F, D \times I)^0, \tag{8.1.6.2}$$

and, by the universal property of the inductive limit of the category fibred over I^0,

$$\mathcal{H}om_{\mathrm{Cart}\,/I}(F, D \times I)^0 \to \mathcal{H}om(\varinjlim_{I^0} F, D)^0. \tag{8.1.6.3}$$

Let $\Pi : F \to \varinjlim_{I^0} F$ be the canonical functor and let Θ denote the composite of these functors.

Proposition 8.1.7. The functor Θ is fully faithful. Its essential image consists of those functors

$$n : \varinjlim_{I^0} F \to D$$

for which $(n \circ \Pi, p) : F \to D \times I$ is a morphism from the total site $D \times I$ to the total site F.

This follows from 7.2.7, 7.2.2, and the universal property of 6.3, together with the criterion 7.4.13.1 for morphisms of total sites.

8.2. Construction of the projective limit when the index category is cofiltered

Lemma 8.2.1. Let I and I' be cofiltered categories, and let $\phi : I' \to I$ be a functor such that $\phi^0 : I'^0 \to I^0$ is cofinal. Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over I, let D be a $\mathcal{U}$-topos, and let $m : D \times I \to F$ be a Cartesian morphism of fibred topoi. If m' is the base change of m to I', then (D, m) is a projective limit of F if and only if (D, m') is a projective limit of F'.

The proof is the expected, somewhat lengthy, verification, or equivalently the same statement for projective limits of pseudo-functors.

Lemma 8.2.2. Let $p : F \to I$ be a $\mathcal{U}$-site fibred over a small cofiltered category, let $\overrightarrow{F} = \varinjlim_{I^0} F$, and let $\pi : F \to \overrightarrow{F}$ be the canonical functor. Put on $\overrightarrow{F}$ the least topology for which π is continuous. For a $\mathcal{U}$-site D and a functor $n : \overrightarrow{F} \to D$, the following conditions are equivalent.

i) n is a morphism of sites $D \to \overrightarrow{F}$.

ii) $n \circ \pi$ is a morphism of sites, where F carries the total topology.

iii) $(n \circ \pi, p) : F \to D \times I$ is a morphism from the total site $D \times I$ to the total site F.

iv) For every object $i \in I$, the composite $F_i \xrightarrow{\alpha_{i!}} F \xrightarrow{n \circ \pi} D$ is a morphism of sites.

The equivalence of (i) and (ii) is 6.6. The equivalence of (iii) and (iv) is 7.4.13. To see that (iii) implies (ii), it is enough to note that the first projection $D \times I \to D$ is a morphism of sites; under the description $(D \times I)^\sim \simeq \mathcal{H}om(I^0, D^\sim)$, its prolongation to sheaves is the functor taking a diagram $\sigma : I^0 \to D^\sim$ to $\varinjlim_{I^0} \sigma(i)$, and filtered colimits are exact. The implication (ii) implies (iv) reduces, by localization and then by passage to associated topoi, to the fact that the canonical section $i \mapsto \alpha_{i!}(e_i)$ becomes final after applying $n \circ \pi$.

Theorem 8.2.3. Let $p : F \to I$ be a $\mathcal{U}$-site fibred over an essentially small cofiltered category.

1) Let $\overrightarrow{F}$ be the site whose underlying category is $\varinjlim_{I^0} F$, equipped with the least topology for which $\pi : F \to \overrightarrow{F}$ is continuous. Then $\overrightarrow{F}$ is a $\mathcal{U}$-site and

$$(\pi, p) : F \to \overrightarrow{F} \times I$$

is a morphism from the fibred site $\overrightarrow{F} \times I$ to the fibred site F.

2) The morphism of fibred topoi

$$\mu : \overrightarrow{F}^{\sim} \times I \to F^{\sim/I}$$

deduced from (π, p) makes $(\overrightarrow{F}^{\sim}, \mu)$ a projective limit of $F^{\sim/I}$.

Thus one may write

$$\varprojlim_{I} F^{\sim} = \overrightarrow{F}^{\sim}. \tag{8.2.4.1}$$

The topology on $\overrightarrow{F}$ is also the least topology making all $F_j \to F \to \overrightarrow{F}$ continuous. The site $\overrightarrow{F}$ is called the *projective-limit site* of the fibred site F.

The proof reduces first to the case where the indexing category is small, by replacing I by a full subcategory whose opposite is cofinal. The first assertion is Lemma 8.2.2 applied to $D = \overrightarrow{F}$ and $n = \mathrm{id}$. For the universal property, one compares the diagram of functors

$$\mathcal{H}om_{\mathrm{top}}(D, \overrightarrow{F}^{\sim}), \quad \mathrm{MorSite}(D, \overrightarrow{F}), \quad \mathcal{H}om(\overrightarrow{F}, D)^0,$$

with their Cartesian analogues over I. The comparison functors are either the standard equivalences for sites and associated topoi, base-change functors, composition with μ or (π, p), or the equivalence supplied by the universal property of $\varinjlim_{I^0} F$. It remains only to compare the essential images of two fully faithful functors into $\mathcal{H}om_{\mathrm{Cart}\,/I}(F, D \times I)^0$; this is exactly Lemma 8.2.2.

8.2.8–8.2.12. Sheaves on the projective limit

Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over I, and suppose that $\varprojlim_I F_i$ exists. Passing from the Cartesian morphism $\mu : \varprojlim_I F_i \times I \to F$ to direct-image functors gives a Cartesian functor

$$\mu_* : \varprojlim_{I} F_i \times I^0 \to F'.$$

Hence there is a canonical functor

$$\Theta : \varprojlim_{I} F_i \to \varprojlim_{I, f_*} F_i = \mathcal{H}om_{\mathrm{Cart}\,/I^0}(I^0, F'). \tag{8.2.8.1}$$

Theorem 8.2.9. Assume that I is essentially small and cofiltered. Then Θ is an equivalence of categories. If I is small and $\varprojlim_I F_i$ is identified with $\overrightarrow{F}^{\sim}$, the composite

$$\overrightarrow{F}^{\sim} \xrightarrow{\Theta} \mathcal{H}om_{\mathrm{Cart}\,/I^0}(I^0, F') \hookrightarrow \mathcal{H}om_{I^0}(I^0, F') \simeq F^{\sim}$$

is precisely the direct image by the morphism of sites $\pi : F \to \overrightarrow{F}$.

The content is often summarized by the formula

$$\varprojlim_{I, f_*} F_i^{\sim} \simeq \varprojlim_{I, f.} F_i^{\sim} \simeq (\varinjlim_{I^0, f^*} F_i)^{\sim}. \tag{8.2.11.1}$$

The theorem says that a sheaf on the projective-limit topos is known by the system of its direct images in the topoi F_i, that is, by a Cartesian section over I^0 of F'. Conversely, a compatible system $(X_i, X_j \simeq f_* X_i)$ comes from an essentially unique sheaf on the limit.

For the proof, one reduces to I small. The functor $\pi : F \to \varinjlim_{I^0} F_i$ is a morphism of sites. Its direct image is fully faithful, with essential image the sheaves on F sending Cartesian morphisms to isomorphisms. Under the equivalence $F^\sim \simeq \mathcal{H}om_{I^0}(I^0, F')$, these sheaves are precisely the Cartesian sections. Moreover, for $X \in \overrightarrow{F}^\sim$, the corresponding section is $j \mapsto \alpha_j^* \pi_* X$, and this is the same as $j \mapsto \mu_{j*} X$. This identifies π_* with the composite displayed above.

8.3. Topology of the projective-limit site: coherent topoi

Throughout 8.3, let $p : F \to I$ be a $\mathcal{U}$-site fibred over an essentially small cofiltered category, and assume that the inverse-image functors commute with fibre products. For every object X of a fibre F_i, choose a set $\mathrm{Cov}(X)$ of covering families satisfying (PT0) and (PT1), generating the topology of F_i, and stable under inverse image along the transition functors.

For an object Y of $\overrightarrow{F}$, define $\mathrm{Cov}(Y)$ to consist of all families $(Y_\alpha \to Y)_\alpha$ obtained, up to isomorphism in $\overrightarrow{F}$, from a covering family $(X_\alpha \to X)_\alpha \in \mathrm{Cov}(X)$ in some fibre F_i.

Proposition 8.3.3. The families belonging to $\mathrm{Cov}(Y)$, for $Y \in \overrightarrow{F}$, generate the topology of the projective-limit site. The assignment $Y \mapsto \mathrm{Cov}(Y)$ satisfies (PT0) and (PT1).

Lemma 8.3.4. Let $i \in I$, and let $n : X' \to X$ be a pullbackable morphism of F_i such that every pullback $f^*(n)$ along $f : j \to i$ is pullbackable. Then $\pi(n)$ is pullbackable in $\overrightarrow{F}$. In particular the canonical functors

$$\pi_j : F_j \hookrightarrow F \xrightarrow{\pi} \overrightarrow{F}$$

commute with fibre products.

Indeed, every morphism to $\pi(X)$ in $\overrightarrow{F}$ may be represented, after inverting a Cartesian morphism, by a morphism in a suitable fibre. Replacing it by the appropriate inverse image of n, the required fibre product is reduced to the fibrewise fibre product. The compatibility of the functors π_j with fibre products then follows from the formula for morphisms in the inductive limit and from the fact that filtered colimits commute with finite fibre products in sets.

The proof of Proposition 8.3.3 is then formal. The preceding lemma gives (PT0). Stability under base change is reduced to the corresponding statement in one fibre, using the fact that π_i commutes with fibre products. Finally, a sheaf for the topology generated by the displayed families is exactly a presheaf M on $\overrightarrow{F}$ for which every $M \circ \pi_i$ is a sheaf on F_i. This is the same as being a sheaf for the least topology making $\pi : F \to \overrightarrow{F}$ continuous.

Proposition 8.3.6. If, for every fibre, $X \mapsto \mathrm{Cov}(X)$ is a pretopology and all covering families in it are finite, then the families $\mathrm{Cov}(Y)$ are finite and form a pretopology on $\overrightarrow{F}$.

Only (PT2), stability under composition, has to be checked. Given a covering $(Y_\alpha \to Y)$ represented in one fibre and coverings of the Y_α, one first replaces the isomorphisms in $\overrightarrow{F}$ by genuine morphisms after pulling back along suitable arrows in I. Since the index category is cofiltered and the covering family is finite, all data may be brought to a single fibre. A further base change reduces to the case where the structural arrow of I is the identity. The final point is the following lemma.

Lemma 8.3.11. Let $j \in I$ and let $m : X' \to X$ be a morphism in F_j such that $\pi(m)$ is an isomorphism. Then there is a morphism $\ell : j' \to j$ such that $\ell^*(m)$ is an isomorphism.

A quasi-inverse to $\pi(m)$ can be represented by a span $X \xleftarrow{s_1} Y_1 \xrightarrow{q_1} X'$ with s_1 Cartesian. After another Cartesian pullback, the equality $\pi(mq_1) = \pi(s_1)$ becomes an equality in a fibre. Factoring through Cartesian morphisms gives a one-sided inverse for a pullback of m; applying the same argument to this inverse gives a two-sided inverse after a further pullback. Since the covering families are finite, one may choose a common pullback for all α, and the verification of (PT2) reduces to the (PT2) axiom in a single fibre.

Theorem 8.3.13. Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over an essentially small cofiltered category. Assume that every fibre F_i is coherent and that, for every arrow $f : i \to j$, the morphism $f. : F_i \to F_j$ is coherent. Then $\varprojlim_I F_i$ is coherent and every canonical morphism

$$\mu_i : \varprojlim_I F_i \to F_i$$

is coherent.

Moreover, if F_{coh} denotes the full subcategory of F consisting of objects coherent in their fibres, then F_{coh} is fibred over I, and $\overrightarrow{F}_{\text{coh}}$ is canonically equivalent to the category of coherent objects of $\varprojlim_I F_i$.

For each i, the site $(F_i)_{\text{coh}}$, with the induced topology, defines F_i. Coherence of the transition morphisms implies that inverse images carry coherent objects to coherent objects, and hence defines a fibred site $F_{\text{coh}} \to I$. By 8.2.3,

$$\varprojlim_I F_i \simeq (\overrightarrow{F}_{\text{coh}})^{\sim}. \tag{8.3.13.1}$$

Finite limits are representable in the coherent sites, and finite covering families form a pretopology. Proposition 8.3.6 shows that the limit site has finite covering families and quasi-compact objects; by 2.4.5 the resulting topos is coherent. The morphisms μ_i are induced by morphisms of coherent sites $(F_i)_{\text{coh}} \to \overrightarrow{F}_{\text{coh}}$, hence are coherent.

For the final assertion, let X be coherent in $(\overrightarrow{F}_{\text{coh}})^{\sim}$. Quasi-compactness gives, after moving to a sufficiently small index, a finite epimorphism $\mu_i(Y') \to X$ with $Y' \in (F_i)_{\text{coh}}$. The corresponding equivalence relation is again coherent. After one more base change it is represented by an actual equivalence relation in $(F_i)_{\text{coh}}$; its quotient is coherent in F_i, and its image is X. Thus every coherent object of the limit comes from $\overrightarrow{F}_{\text{coh}}$.

Corollary 8.3.14. Under the hypotheses of 8.3.13, let $F \to I$ and $G \to I$ be coherent fibred topoi with coherent transition morphisms, and let $m : F \to G$ be a Cartesian morphism such that every fibre morphism $m_i : F_i \to G_i$ is coherent. Then the induced morphism $\overleftarrow{m} : \overleftarrow{F} \to \overleftarrow{G}$ is coherent.

This follows by applying 8.3.13 to the fibred sites of coherent objects and using 3.3.

Exercise 8.3.15. In the notation of 8.3.13, show that if the F_i are algebraic and the transition morphisms are coherent, then $\varprojlim_I F_i$ is algebraic and $(\varprojlim_I F_i)_{\text{coh}} \simeq \overrightarrow{F}_{\text{coh}}$.

8.4. Example: local topoi

Let $X = (X_i)_{i \in I}$ be a topos fibred over I. Extending it to pro-objects, one obtains a fibred topos over $\text{Pro}(I)$ by

$$X_\alpha = \varprojlim_{i \in I_\alpha} X_i \tag{8.4.1.1}$$

for every pro-object $\alpha = (i_\alpha)$ of I.

Let E be a topos and let $\mathrm{Fl}(E) \to E$ be the fibred topos of 7.3.1. Extending it to $\mathrm{Pro}(E)$, and restricting along the fully faithful functor $\mathrm{Point}(E) \to \mathrm{Pro}(E)$, gives a fibred topos

$$\mathrm{Loc}(E) \to \mathrm{Point}(E).$$

The fibre at a point p is called the *localization of E at p* and is denoted $\mathrm{Loc}_p(E)$. By definition,

$$\mathrm{Loc}_p(E) = \varprojlim_{X \in \mathrm{Vois}(p)} E_{/X}, \tag{8.4.2.1}$$

where $\mathrm{Vois}(p)$ is the category of neighbourhoods of p.

If $m : p' \to p$ is a morphism of points of E, the universal property gives a point $\theta_p(m)$ of $\mathrm{Loc}_p(E)$, hence a functor

$$\theta_p : \mathrm{Point}(E)_{/p} \to \mathrm{Point}(\mathrm{Loc}_p(E)). \tag{8.4.3.1}$$

This functor is an equivalence of categories. Therefore $\mathrm{Point}(\mathrm{Loc}_p(E))$ is canonically equivalent to the category of generizations of p. There is also a canonical equivalence

$$\mathrm{Loc}_{\theta_p(m)}(\mathrm{Loc}_p(E)) \xrightarrow{\sim} \mathrm{Loc}_{p'}(E),$$

compatible with the canonical morphisms to $\mathrm{Loc}_p(E)$.

These localized topoi are mainly useful for topoi arising in algebraic geometry, or at least for topoi with finiteness properties. For a separated topological space, the localizations are punctual topoi, and the situation is essentially trivial. If $E = \mathrm{Top}(X)$ is the Zariski topos of a scheme X, if $x \in X$, and if $Y = \mathrm{Spec}(\mathcal{O}_{X,x})$, then

$$\mathrm{Loc}_x(E) = \mathrm{Top}(Y).$$

Étale examples occur later in Exposé VIII.

If E is locally coherent, the topoi $\mathrm{Loc}_p(E)$ are coherent; it is enough in 8.4.2 to restrict to algebraic coherent neighbourhoods and then apply 8.3.13. If $m : p' \to p$ is a morphism of points, the induced morphism $\mathrm{Loc}_{p'}(E) \to \mathrm{Loc}_p(E)$ is coherent.

A topos X is called *local* if the global-sections functor $\Gamma(X, -)$ is a fibre functor. The corresponding point is called the *centre* of the local topos. When E is locally coherent, the localized topoi $\mathrm{Loc}_p(E)$ are local; the centre of $\mathrm{Loc}_p(E)$ is canonically $\theta_p(\mathrm{id}_p)$, and its image in E is p. The topos $\mathrm{Loc}_p(E)$, with its canonical morphism to E, solves the 2-universal problem of mapping local topoi into E with the centre lying over p.

Several natural questions remain. If X is local, the final object has a maximal open $U \neq e$; is the closed complement, which is again local and has no nontrivial open, necessarily punctual? If X is obtained by gluing a topos U with the punctual topos through a functor $\Phi : U \to \mathbf{Set}$, what conditions on Φ make X local?

8.5. Sheaves on a filtered projective limit of topoi

Let $F = (F_i)_{i \in I}$ be a $\mathcal{U}$-topos fibred over a small cofiltered category, and choose a cleavage and an opcleavage. Let

$$\overleftarrow{F} = \varprojlim_I F_i, \qquad \mu_i : \overleftarrow{F} \to F_i. \tag{8.5.1.1}$$

Let $\mathrm{Top}(F)$ be the total topos. The canonical functor $F \to \overrightarrow{F}$ defines a morphism

$$Q : \overleftarrow{F} \to \mathrm{Top}(F). \tag{8.5.1.2}$$

Under the identifications

$$\overleftarrow{F} \simeq \mathcal{H}om_{\mathrm{Cart}\,/I^0}(I^0, F'), \qquad \mathrm{Top}(F) \simeq \mathcal{H}om_{I^0}(I^0, F'),$$

the morphism Q is the inclusion of Cartesian sections into all sections. Thus, for $M \in \overleftarrow{F}$,

$$Q_*M = (i \mapsto \mu_{i*}M). \tag{8.5.1.3}$$

For each i there is a canonical morphism $\alpha_i : F_i \to \mathrm{Top}(F)$, and although the triangle $\overleftarrow{F} \to F_i \to \mathrm{Top}(F)$ need not commute, one always has the canonical isomorphism

$$\alpha_i^* Q_* \simeq \mu_{i*}. \tag{8.5.1.6}$$

Proposition 8.5.2. For an object $(j \mapsto M_j)$ of $\mathrm{Top}(F)$, there is a functorial isomorphism

$$Q^*(j \mapsto M_j) \simeq \varinjlim_{I^0} \mu_j^* M_j. \tag{8.5.2.1}$$

An object of $\mathrm{Top}(F)$ consists of objects $M_j \in F_j$ and transition maps

$$\beta_f : M_j \to f_* M_i \tag{8.5.2.2}$$

or, by adjunction, $\beta'_f : f^*M_j \to M_i$, satisfying the usual cocycle condition. Combining β'_f with the transition isomorphisms $b_f : f.\mu_i \simeq \mu_j$ gives morphisms

$$t_f : \mu_j^* M_j \to \mu_i^* M_i, \tag{8.5.2.6}$$

with $t_f t_g = t_{gf}$. Hence the displayed filtered colimit is defined. For $N \in \overleftarrow{F}$, adjunction gives

$$\mathrm{Hom}(\varinjlim_{I^0} \mu_i^* M_i, N) \simeq \varprojlim_{I^0} \mathrm{Hom}_{F_i}(M_i, \mu_{i*}N),$$

where the right hand side is precisely the set of morphisms from the section $(i \mapsto M_i)$ to Q_*N. This proves the proposition.

Proposition 8.5.3. Assume that for every arrow $f : i \to j$ in I, the functor $f_* : F_i \to F_j$ commutes with small filtered colimits. Then, for every section $(i \mapsto M_i)$ of F',

$$Q_*Q^*(i \mapsto M_i) \simeq (i \mapsto \varinjlim_{f:j \to i} f_* M_j). \tag{8.5.3.1}$$

The transition maps in the right hand side are induced by the maps β_f and the coherence isomorphisms $c_{f,g}$. Under the stated commutation hypothesis the resulting section is Cartesian, and the natural map from $(i \mapsto M_i)$ to it is universal among maps to Cartesian sections.

Corollary 8.5.4. Under the hypotheses of 8.5.3, the functor Q_* commutes with small filtered colimits.

This follows by writing an object N as Q^*Q_*N, using the exactness of Q^*, computing colimits in $\mathrm{Top}(F)$ fibre by fibre, and then applying 8.5.3.

Corollary 8.5.5. Under the same hypotheses, for every $i \in I$ and every $M_i \in F_i$,

$$\mu_{i*}\mu_i^* M_i \simeq \varinjlim_{f:j \to i} f_* f^* M_i. \tag{8.5.5.1}$$

After replacing I by $I_{/i}$, one may suppose that i is final. Then μ_i factors through Q and the retraction of $\mathrm{Top}(F)$ to the final fibre, and the formula is an instance of 8.5.3.

Corollary 8.5.6. The functors $\mu_{i*} : \overleftarrow{F} \to F_i$ commute with filtered colimits.

Indeed μ_{i*} is the composite of Q_* with restriction to the fibre i.

Corollary 8.5.7. Assume 8.5.3. Let $i \in I$ and let $X \in F_i$ be such that $\mathrm{Hom}(X, -)$ commutes with filtered colimits on F_i. Then $\mathrm{Hom}(\mu_i^* X, -)$ commutes with filtered colimits on $\overleftarrow{F}$, and for every $M_i \in F_i$,

$$\mathrm{Hom}(\mu_i^* X, \mu_i^* M_i) \simeq \varinjlim_{f : j \to i} \mathrm{Hom}(f^* X, f^* M_i). \tag{8.5.7.1}$$

For every section $(j \mapsto M_j)$ of $\mathrm{Top}(F)$,

$$\mathrm{Hom}(\mu_i^* X, Q^*(j \mapsto M_j)) \simeq \varinjlim_{f : j \to i} \mathrm{Hom}(f^* X, M_j). \tag{8.5.7.2}$$

The functor $\mathrm{Hom}(\mu_i^* X, -)$ is $\mathrm{Hom}(X, \mu_{i*} -)$; the preceding corollaries and the hypothesis on X prove the colimit statement and the formulas.

Proposition 8.5.8. Let $G \to I$ be another $\mathcal{U}$-topos fibred over I, and let $m : F \to G$ be a Cartesian morphism of fibred topoi. The square

$$\begin{tikzcd} \overleftarrow{F} \arrow[r, "Q"] \arrow[d, "\overleftarrow{m}"'] & \mathrm{Top}(F) \arrow[d, "m^{\sim}"] \\ \overleftarrow{G} \arrow[r, "Q"] & \mathrm{Top}(G) \end{tikzcd} \tag{8.5.8.1}$$

commutes up to canonical isomorphism. For every $N \in \overleftarrow{F}$,

$$\overleftarrow{m}_* N \simeq \varinjlim_{I^0} \mu_j^* m_{j*} \mu_{j*} N. \tag{8.5.8.2}$$

The commutativity is immediate from the definitions. Since $N \simeq Q^* Q_* N$, the formula follows from 8.5.2 applied to the section $j \mapsto m_{j*} \mu_{j*} N$.

Proposition 8.5.9. Assume, in addition to the hypotheses of 8.5.8, that every m_{i*} and every transition functor f_* on F commute with filtered colimits. Then $\overleftarrow{m}_*$ commutes with filtered colimits and, for every section $(i \mapsto M_i)$ of $\mathrm{Top}(F)$,

$$\overleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{I^0} \mu_j^* m_{j*} M_j. \tag{8.5.9.1}$$

The proof starts from 8.5.8 and 8.5.3:

$$\overleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{i} \mu_i^* m_{i*} \big(\varinjlim_{f : j \to i} f_* M_j \big).$$

Using the commutation of m_{i*} with filtered colimits and the canonical isomorphisms $m_{i*} f_* \simeq f_* m_{j*}$, this becomes a colimit over the category of arrows of I. The functor from I to the arrow category which sends i to id_i is cofinal on opposites, giving the displayed formula.

Corollary 8.5.10. Under the conditions of 8.5.9, for every $i \in I$ and $M_i \in F_i$,

$$\overleftarrow{m}_* \mu_i^* M_i \simeq \varinjlim_{f:j \to i} \mu_j^* m_{j*} f^* M_i. \tag{8.5.10.1}$$

Remark 8.5.11. Recall that the direct image by a coherent morphism between coherent topoi commutes with filtered colimits. Thus 8.5.3–8.5.10 apply in particular to coherent fibred topoi and coherent Cartesian morphisms between them.

8.6. Ringed fibred $\mathcal{U}$-topoi

A $\mathcal{U}$-topos fibred over I is *ringed* if it is equipped with a sheaf of rings on its total site. After choosing a cleavage and an opcleavage, this is the same as giving rings A_i in the fibres F_i, together with morphisms $A_j \to f_* A_i$ for every arrow $f : i \to j$, subject to the usual compatibility conditions. Thus the transition morphisms $f. : F_i \to F_j$ become morphisms of ringed topoi, and the data amount to a pseudo-functor from I to the 2-category of ringed topoi. Conversely, such a pseudo-functor reconstructs the ringed fibred topos.

The notion of projective limit extends immediately to ringed $\mathcal{U}$-topoi. If the underlying fibred topos has a projective limit $\varprojlim_I F_i$, and if the inductive system of rings $i \mapsto \mu_i^* A_i$ has a colimit in the category of rings of the limit topos, then

$$(\varprojlim_I F_i,\ \varinjlim_{I^0} \mu_i^* A_i)$$

is the corresponding projective limit in ringed topoi.

The formulas of 8.5 remain valid for abelian sheaves, and for modules provided inverse images are interpreted as inverse images of modules. A ringed fibred topos $(F, A) \to I$ is said to be flat on the right, respectively on the left, if for every $f : i \to j$ the morphism $f. : (F_i, A_i) \to (F_j, A_j)$ is flat on the right, respectively on the left. If I is essentially small, then

$$Q : (\varprojlim_I F_i, \varinjlim_{I^0} \mu_i^* A_i) \to (\mathrm{Top}(F), A)$$

is flat on the right, respectively on the left, without any flatness assumption on the transition morphisms. If the ringed fibred topos is flat on the right, respectively left, then each μ_i is flat on the same side. Similarly, a Cartesian morphism of ringed fibred topoi whose fibre morphisms are flat on one side induces a projective-limit morphism flat on that side.

8.7. Cohomology of sheaves on a projective limit of topoi

In 8.7 let I be a small cofiltered category, let $(F, A) \to I$ and $(G, B) \to I$ be ringed $\mathcal{U}$-topoi fibred over I, and let

$$m : (F, A) \to (G, B)$$

be a morphism of ringed fibred topoi. Assume that, for every arrow $f : i \to j$, the derived functors $R^n f_*$ on modules commute with filtered colimits, and that for every object i, the derived functors $R^n m_{i*}$ commute with filtered colimits. This is the case, for example, when the fibres and the relevant morphisms are coherent. Write

$$\overleftarrow{A} = \varinjlim \mu_i^* A_i, \qquad \overleftarrow{B} = \varinjlim \mu_i^* B_i.$$

For modules, write μ_i^{-1} for inverse image as abelian sheaves, and reserve μ_i^* for inverse image of modules.

Lemma 8.7.2. If $(j \mapsto M_j)$ is an injective A-module of $\mathrm{Top}(F)$, then $Q^*(j \mapsto M_j)$ is acyclic for $\overleftarrow{m}_*$, and every M_i is flasque.

The flasqueness follows by restricting the flasque module $(j \mapsto M_j)$ to the localized fibred topos over $I_{/i}$, and then using the retraction from the total topos to the fibre. For acyclicity, put $N' = Q^*(j \mapsto M_j)$. Formula 8.5.3 gives

$$\mu_{i*}N' \simeq \varinjlim_{f:j\to i} f_*M_j.$$

Using the hypotheses, N' has the property that each $\mu_{i*}N'$ is acyclic for m_{i*} and for all transition functors. This property is stable under the standard dévissage criterion for acyclicity: in an exact sequence $0 \to N' \to N \to N'' \to 0$, the cokernel of $Q_*N' \to Q_*N$ is again a Cartesian section and hence is Q_* of an object of the projective limit; then 8.5.8, exactness of μ_j^{-1}, and exactness of filtered colimits give the required epimorphism after applying $\overleftarrow{m}_*$.

Theorem 8.7.3. Under the hypotheses of 8.7.1, for every $n \in \mathbb{N}$, the functor $R^n\overleftarrow{m}_*$ commutes with filtered colimits. Moreover, for every A-module section $(j \mapsto M_j)$,

$$R^n\overleftarrow{m}_*\, Q^*(j \mapsto M_j) \simeq \varinjlim_{I^0} \mu_j^* R^n m_{j*} M_j. \tag{8.7.3.1}$$

Since $Q^{-1}A = \overleftarrow{A}$, inverse image for modules agrees here with inverse image for abelian sheaves followed by extension of scalars. Lemma 8.7.2 computes the derived functor on a resolution by injective sections, giving the formula; commutation with filtered colimits follows at once.

Corollary 8.7.4. For every $n \in \mathbb{Z}$ and every module N on $\overleftarrow{F}$,

$$R^n\overleftarrow{m}_*N \simeq \varinjlim_{I^0} \mu_j^* R^n m_{j*}\mu_{j*}N. \tag{8.7.4.1}$$

Indeed $N \simeq Q^*(j \mapsto \mu_{j*}N)$.

Corollary 8.7.5. For every object $i \in I$, every integer n, and every A_i-module M_i,

$$R^n\overleftarrow{m}_*\mu_i^*M_i \simeq \varinjlim_{f:j\to i} \mu_j^* R^n m_{j*} f^* M_i. \tag{8.7.5.1}$$

After base change along $I_{/i} \to I$, one may suppose that i is final; then $\mu_i^*M_i$ is the inverse image under Q of the section $(f : j \to i) \mapsto f^*M_i$.

Corollary 8.7.6. For every $i \in I$ and every integer n,

$$R^n\mu_{i*}Q^*(j \mapsto M_j) \simeq \varinjlim_{f:j\to i} R^n f_* M_j, \tag{8.7.6.1}$$

$$R^n\mu_{i*}N \simeq \varinjlim_{f:j\to i} R^n f_*\mu_{j*}N, \tag{8.7.6.2}$$

and

$$R^n\mu_{i*}\mu_i^*M_i \simeq \varinjlim_{f:j\to i} R^n f_* f^* M_i. \tag{8.7.6.3}$$

The functors $R^n\mu_{i*}$ commute with filtered colimits.

This is obtained by applying 8.7.3–8.7.5 to the constant fibred topos with fibre F_i, after the base change $I_{/i} \to I$.

Corollary 8.7.7. Let $i \in I$ and let $X \in F_i$ be such that all functors $H^n(X, -)$ commute with filtered colimits. Then, for every n, there is a canonical isomorphism, functorial in the A_i-module M_i,

$$H^n(\mu_i^* X, \mu_i^* M_i) \simeq \varinjlim_{f: j \to i} H^n(f^* X, f^* M_i), \tag{8.7.7.1}$$

and the functors $H^n(\mu_i^* X, -)$ commute with filtered colimits. In particular, if the functors $H^n(F_i, -)$ commute with filtered colimits, then the functors $H^n(\overleftarrow{F}, -)$ commute with filtered colimits, and

$$H^n(\overleftarrow{F}, \mu_i^* M_i) \simeq \varinjlim_{f: j \to i} H^n(F_j, f^* M_i). \tag{8.7.7.2}$$

The proof compares the two Cartan–Leray spectral sequences

$$E_2'^{p,q} = \varinjlim_{f: j \to i} H^p(X, R^q f_* f^* M_i) \Rightarrow \varinjlim_{f: j \to i} H^{p+q}(f^* X, f^* M_i)$$

and

$$E_2''^{p,q} = H^p(X, R^q \mu_{i*} \mu_i^* M_i) \Rightarrow H^{p+q}(\mu_i^* X, \mu_i^* M_i).$$

Formula 8.7.6 identifies the E_2-terms, so the abutments agree. The filtered-colimit assertion follows from the analogous assertions for $H^n(X, -)$ and for $R^n \mu_{i*}$.

Remark 8.7.8. When the F_j are coherent and the transition morphisms are coherent, the hypotheses made on X and F_i in 8.7.7 are satisfied for coherent X and coherent F_i, by 5.2. In the same situation $\mu_i^* X$, respectively $\overleftarrow{F}$, is coherent by 8.3.13.

Corollary 8.7.9. Let $i \in I$, let M_i be a left A_i-module, and let L_i be a right A_i-module admitting a resolution

$$P_{i,k} \to P_{i,k-1} \to \cdots \to P_{i,0} \to L_i \to 0$$

where each $P_{i,k}$ is a finite direct sum of modules of the form $A_{i|X}$, with X satisfying the hypotheses of 8.7.7. If, in addition to the hypotheses of 8.7.1, the fibred topos F is flat on the right, then, for $n \leq k-1$, there are canonical isomorphisms

$$\mathrm{Ext}^n_{\overleftarrow{A}}(\overleftarrow{F}, \mu_i^* L_i, \mu_i^* M_i) \simeq \varinjlim_{f: j \to i} \mathrm{Ext}^n_{A_j}(F_j, f^* L_i, f^* M_i). \tag{8.7.9.1}$$

If $P_{i,\bullet}$ is the chosen resolution of L_i, flatness gives resolutions $f^* P_{i,\bullet}$ of $f^* L_i$ and $\mu_i^* P_{i,\bullet}$ of $\mu_i^* L_i$. The two spectral sequences obtained from these resolutions converge to the two sides of 8.7.9.1; on the E_1-page the comparison is the isomorphism 8.7.7.1, hence it is an isomorphism on abutments.

9. Appendix. Criterion for the existence of points

by P. Deligne

Proposition 9.0. Every locally coherent topos S has enough points.

Since the assertion is local on S, one may suppose that S is defined by a site $\mathscr{S}$ in which finite projective limits are representable and in which every covering family admits a finite subcover. It is enough to show that if a morphism of sheaves $u : \mathcal{F} \to \mathcal{G}$ is not a monomorphism, then there is a point x of $\mathscr{S}$ such that u_x is not injective. Thus there are an object $U \in \mathscr{S}$ and two sections $s, s' \in \mathcal{F}(U)$, with $s \neq s'$, and with $u(s) = u(s')$. Replacing $\mathscr{S}$ by $\mathscr{S}/U$, the assertion reduces to the following lemma.

Lemma 9.1. Let s and s' be two distinct global sections of a sheaf $\mathcal{F}$ on a site $\mathscr{S}$ satisfying the preceding hypotheses. Then there is a point x of $\mathscr{S}$ such that $s_x \neq s'_x$.

Let $P = (U_i)_{i \in I}$ be a projective system in $\mathscr{S}$, indexed by a filtered ordered set. For every sheaf $\mathcal{H}$ on $\mathscr{S}$, set

$$P(\mathcal{H}) = \varinjlim_i \mathcal{H}(U_i),$$

and for every object V of $\mathscr{S}$, set

$$P(V) = \varinjlim_i \mathrm{Hom}(U_i, V).$$

These functors commute with fibre products. If $P(V)$ sends coverings to surjective families of sets, then it defines a morphism of sites from the punctual topos to $\mathscr{S}$, and $P(\mathcal{H})$ is the associated fibre functor.

If $P = (U_i)_{i \in I}$ and $Q = (V_j)_{j \in J}$ are two such projective systems, we say that Q *refines* P if I is an ordered subset of J and P is the restriction of Q to I. There are then natural maps $P(\mathcal{H}) \to Q(\mathcal{H})$ and $P(V) \to Q(V)$.

For such a P, write s_P and s'_P for the images of s and s' in $P(\mathcal{F})$. Lemma 9.1 follows from the next lemma, applied to the projective system indexed by $\{0\}$ and reduced to the final object of $\mathscr{S}$.

Lemma 9.2. For every P with $s_P \neq s'_P$, there is a refinement Q of P, with $s_Q \neq s'_Q$, such that, for every covering $(V_i \to V)$ and every element $f \in Q(V)$, the element f lies in the image of one of the maps $Q(V_i) \to Q(V)$.

We first prove a finite form.

Lemma 9.3. Let $P = (U_i)_{i \in I}$ satisfy the hypotheses of Lemma 9.2. Let $(V_k \to V)_k$ be a finite covering in $\mathscr{S}$, and let $f \in P(V)$. There is a refinement Q of P, with $s_Q \neq s'_Q$, such that the image of f in $Q(V)$ lies in the image of one of the $Q(V_k)$.

Choose $i_0 \in I$ and a morphism $f' : U_{i_0} \to V$ representing f. For $i \geq i_0$, put

$$U_{i,k} = U_i \times_V V_k.$$

Since (V_k) covers V, the family $(U_{i,k} \to U_i)_k$ covers U_i. Therefore

$$\mathcal{F}(U_i) \to \prod_k \mathcal{F}(U_{i,k})$$

is injective, and after passing to the filtered colimit, finite products commuting with filtered colimits, one gets an injection

$$P(\mathcal{F}) \to \prod_k \varinjlim_i \mathcal{F}(U_{i,k}).$$

Hence for some k, the images of s_P and s'_P are distinct in $\varinjlim_i \mathcal{F}(U_{i,k})$. Let $I_0 = \{i \in I \mid i \geq i_0\}$, and form an ordered set $J = I \amalg I_0$ by placing the new copy of I_0 beneath the old terms in the evident way. Index by J the objects U_i and $U_{i,k}$. The resulting system Q refines P, still separates s and s', and the image of f in $Q(V)$ comes from $Q(V_k)$.

Lemma 9.4. Let P satisfy the hypotheses of Lemma 9.2. There is a refinement Q of P, with $s_Q \neq s'_Q$, such that, for every finite covering $(V_k \to V)$ and every $f \in P(V)$, the image of f in $Q(V)$ lies in the image of one of the $Q(V_k)$.

Let E be the set of triples consisting of an object V, a finite covering (V_k) of V, and an element $f \in P(V)$. Choose a well-ordering on E, and adjoin a largest element ∞. By transfinite induction construct refinements Q_e of P, with $s_{Q_e} \neq s'_{Q_e}$, such that Q_e refines $Q_{e'}$ for $e' \leq e$, and such that the condition corresponding to $e = (V, (V_k), f)$ is satisfied at stage e. At successor stages apply Lemma 9.3; at limit stages take the union of the preceding systems. Because filtered colimits are used in the definition of $Q(\mathcal{F})$, the inequality of the two sections is preserved at limit stages. Then Q_∞ proves the lemma.

Lemma 9.4 is now iterated. Starting with $Q_0 = P$, construct refinements Q_{n+1} of Q_n such that $s_{Q_n} \neq s'_{Q_n}$ and such that every covering condition for elements of $Q_n(V)$ is solved in Q_{n+1}. Let Q be the projective system obtained by taking the union of all the index sets. Then Q satisfies Lemma 9.2. The associated functor is therefore a point of $\mathscr{S}$, and by construction it separates s and s'. This proves Lemma 9.1 and hence Proposition 9.0.

The construction also gives the following cardinal refinement.

Corollary 9.5. Let $\mathscr{S}$ be a site in which fibre products are representable and in which every covering admits a finite subcover. Let c be the cardinal $\sup(\aleph_0, \operatorname{card}(\operatorname{Fl}\mathscr{S}))$. Then there is a very dense set X of points of $\mathscr{S}$ such that

i) $\operatorname{card}(X) \leq 2^c$;

ii) if $x \in X$ and $U \in \operatorname{Ob}\mathscr{S}$, then $\operatorname{card}(U_x) \leq c$.

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